

Data Types

Data Type	Description	JSON Schema Type	Examples
Integer	A positive or negative whole number (i.e., a number that can be written without a fractional part).	integer	3, 19, -4
Numeric	A number that may include a fractional part with optional leading sign and optional exponent (engineering notation).	number	3.43, 0, -4, 1.03e4
Boolean	True or false.	boolean	true, false
String	A sequence of characters of any length using any (specified) character set.	string	Indirect evaporative cooler
ID	A referencable identification for a data group and sequence of characters of any length using any (specified) character set.	string	AHU-01
Reference	A reference to an ID of a data group.	string	AHU-01
Null	Indicator that no value is provided. Only used in combination with other data types, e.g., 'Number/Null'.	null	null

String Types

String Type	Description	JSON Schema Pattern	Examples
Timestamp	Date with UTC time formatted per ISO 8601 (ISO 2004)	[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}Z	2016-06-29T14:35Z
Version	Version identifier in the form major.minor.patch as defined by SemVer[@SemVer]	(0 [1-9][0-9]).(0 [1-9][0-9]).(0 [1-9][0-9])(?:(?:(?![1-9][0-9] [0-9][a-zA-Z-])[0-9a-zA-Z-])?(?:(?![1-9][0-9] [0-9][a-zA-Z-])[0-9a-zA-Z-]))?(?:(?+[0-9a-zA-Z-]+(?![0-9a-zA-Z-]+)*)?)?	1.1.3, 1.2.0-beta-92

AirEconomizerOptions

Enumerator	Description	Notes
FIXED_FRACTION	Fixed Fraction	
TEMPERATURE	Dry-bulb temperature	
ENTHALPY	Enthalpy	
DIFFERENTIAL_TEMPERATURE	Differential dry-bulb temperature	
DIFFERENTIAL_ENTHALPY	Differential enthalpy	
OTHER	Other	

ApplicationTemperatureOptions

Enumerator	Description	Notes
MEDIUM	Medium temperature	3.3 C +/- 1.1 C (38 F +/- 2 F)
LOW	Low temperature	-17.8 C +/- 1.1 C (0 F +/- 2 F)
VERY_LOW	Very low	-26.1 C +/- 1.1 C (-15 F +/- 2 F). This corresponds to the ice cream category in AHRI 1200
OTHER	Other	

BoilerCombustionOptions

Enumerator	Description	Notes
NATURAL	Natural	
FORCED	Forced	

BoilerEfficiencyMetricOptions

Enumerator	Description	Notes
ANNUAL_FUEL_UTILIZATION	Annual fuel utilization efficiency	
THERMAL	Thermal efficiency	

Enumerator	Description	Notes
COMBUSTION	Combustion efficiency	

ChillerCompressorOptions

Enumerator	Description	Notes
SCREW	Screw	
CENTRIFUGAL	Centrifugal	
RECIPROCATING	Reciprocating	
SCROLL	Scroll	
POSITIVE_DISPLACEMENT	Positive displacement	Only use the POSITIVE_DISPLACEMENT enumerator when the chiller is positive displacement but it is not known whether it is RECIPROCATING, SCREW, or SCROLL.
SINGLE_EFFECT_INDIRECT_FIRED_ABSORPTION	Single-effect indirect-fired absorption	
DOUBLE_EFFECT_INDIRECT_FIRED_ABSORPTION	Double-effect indirect-fired absorption	
SINGLE_EFFECT_DIRECT_FIRED_ABSORPTION	Single-effect direct-fired absorption	
DOUBLE_EFFECT_DIRECT_FIRED_ABSORPTION	Double-effect direct-fired absorption	
OTHER	Other	

ChillerEfficiencyMetricOptions

Enumerator	Description	Notes
FULL_LOAD_EFFICIENCY_RATED	Full Load Efficiency expressed as a coefficient of performance (COP) at rated conditions	This should correspond with the rated_entering_condenser_temperature and rated_leaving_evaporator_temperature.

Enumerator	Description	Notes
FULL_LOAD_EFFICIENCY_DESIGN	Full Load Efficiency expressed as a coefficient of performance (COP) at design conditions	This should correspond with the design_entering_condenser_temperature and design_leaving_evaporator_temperature.
INTEGRATED_PART_LOAD_VALUE	Integrated part load value efficiency expressed as a coefficient of performance (COP)	This should correspond with the AHRI rated conditions for the equipment, see AHRI 550/590
NONSTANDARD_PART_LOAD_VALUE	Nonstandard part load value efficiency expressed as a coefficient of performance (COP)	This should correspond with the applications conditions defined by the user other than AHRI rated conditions, see AHRI 550/590
OTHER	Other part load efficiency metric	

ChillerFunctionOptions

Enumerator	Description	Notes
CHILLED_WATER_ONLY	Chilled water only	Chiller is being used as a traditional chiller to only produce chilled water.
HOT_WATER_ONLY	Hot water only	Chiller is being used exclusively as a heat pump instead of a boiler.
CHILLED_OR_HOT_WATER	Chilled or hot water	Chiller is reversible and can produce either chilled or hot water.
OTHER	Other	

CommonConstructionClassificationOptions

Enumerator	Description	Notes
METAL_BUILDING	Metal building	
WOOD_FRAMED	Wood-framed	
INSULATION_ENTIRELY_ABOVE_DECK	Insulation entirely above deck	
ATTIC	Attic	
BELOW_GRADE_WALL	Below-grade wall	

Enumerator	Description	Notes
STEEL_JOIST	Steel joist	
SLAB_ON_GRADE	Slab-on-grade	
OTHER	Other	

CommonOutputSchemaOptions

Enumerator	Description	Notes
OTHER	other	

CommonRulesetModelOptions

Enumerator	Description	Notes
USER	The building model as described without consideration of the ruleset	
PROPOSED	The proposed building model	The proposed building model typically transformed from the USER model following specific changes for the ruleset

ComponentLocationOptions

Enumerator	Description	Notes
CONDITIONED	Conditioned	The component location has thermal properties similar to those of a conditioned location.
SEMICONDITIONED	Semiconditioned	The component location has thermal properties similar to those of a semi-conditioned location. It is usually kept warm enough to prevent freezing and often only cooled by passive ventilators.
OUTSIDE	Outside	The component location has thermal properties similar to the outdoors.
GARAGE	Garage	The component location has thermal properties similar to those of a garage. It may or may not have passive ventilators and share an insulated wall with a conditioned zone.

Enumerator	Description	Notes
ATTIC	Attic	The component location has thermal properties similar to those of an attic. It may have passive ventilators or not, is often warmer during summer, especially on sunny days, and may share an insulated ceiling with a conditioned zone.
CRAWL_SPACE	Crawl space	The component location has thermal properties similar to those of a crawl space beneath a building. It may or may not have passive ventilators and share an insulated floor with a conditioned zone.
UNDERGROUND	Underground	The component location has thermal properties similar to those of an underground area.
UNCONDITIONED	Unconditioned	The component location has thermal properties similar to those of an unconditioned area other than one of the other enumerations in this list.
OTHER	Other	The component location has thermal properties similar to those of an area not otherwise described by a different enumeration.

ConditioningOptions

Enumerator	Description	Notes
HEATED_AND_COOLED	Heated and cooled	
HEATED_ONLY	Heated only	
SEMIHEATED	Semiheated	
UNCONDITIONED	Unconditioned	

CoolingDesignDayOptions

Enumerator	Description	Notes
COOLING_0_4	Cooling design day 0.4% annual cumulative frequency of occurrence	
COOLING_1_0	Cooling design day 1.0% annual cumulative frequency of occurrence	
COOLING_2_0	Cooling design day 2.0% annual cumulative frequency of occurrence	

CoolingMetricOptions

Enumerator	Description	Notes
FULL_LOAD_COEFFICIENT_OF_PERFORMANCE	Full load efficiency expressed as a coefficient of performance at 35C/95F rated conditions	
FULL_LOAD_COEFFICIENT_OF_PERFORMANCE_NO_FAN	Full load efficiency expressed as a coefficient of performance not including indoor fan power at 35C/95F rated conditions	Describes the efficiency not including the indoor supply fan. The outdoor evaporator fan power is generally included.
ENERGY_EFFICIENCY_RATIO	EER - Energy efficiency ratio	
SEASONAL_ENERGY_EFFICIENCY_RATIO	SEER - Seasonal energy efficiency ratio	
SEASONAL_ENERGY_EFFICIENCY_RATIO_2	SEER2 - Seasonal energy efficiency ratio 2	
INTEGRATED_ENERGY_EFFICIENCY_RATIO	IEER - Integrated energy efficiency ratio	
INTEGRATED_PART_LOAD_VALUE	IPLV - Integrated part load value	
COMBINED_ENERGY_EFFICIENCY_RATIO	CEER - Combined energy efficiency ratio	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_WATER_LOOP	Coefficient of performance at entering temperature of 30C/86F rated conditions for water loops per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_WATER_LOOP_NO_FAN	Coefficient of performance not including fan at entering temperature of 30C/86F rated conditions for water loops per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_GROUND_WATER	Coefficient of performance at entering temperature of 15C/59F rated conditions for ground water per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_GROUND_WATER_NO_FAN	Coefficient of performance not including fan at entering temperature of 15C/59F rated conditions for ground water per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.

Enumerator	Description	Notes
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_AIR_GROUND_LOOP	Coefficient of performance at entering temperature of 25C/77F rated conditions for ground water per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_AIR_GROUND_LOOP_NO_FAN	Coefficient of performance not including fan at entering temperature of 25C/77F rated conditions for ground water per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.
COEFFICIENT_OF_PERFORMANCE_WATER_TO_WATER_WATER_LOOP	Coefficient of performance at entering temperature of 30C/86F rated conditions for water loops per ISO 13256-2	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_WATER_GROUND_WATER	Coefficient of performance at entering temperature of 15C/59F rated conditions for ground water per ISO 13256-2	
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_WATER_GROUND_LOOP	Coefficient of performance at entering temperature of 25C/77F rated conditions for ground water per ISO 13256-2	
NONE	None	
OTHER	Other	

CoolingSourceOptions

Enumerator	Description	Notes
CHILLED_WATER	Chilled water	
NONE	None	
OTHER	Other	

CoolingSystemOptions

Enumerator	Description	Notes
DIRECT_EXPANSION	Direct expansion	
FLUID_LOOP	Fluid loop	

Enumerator	Description	Notes
NON_MECHANICAL	Non-mechanical	
NONE	None	
OTHER	Other	

DayOfWeekOptions

Enumerator	Description	Notes
SUNDAY	Sunday	
MONDAY	Monday	
TUESDAY	Tuesday	
WEDNESDAY	Wednesday	
THURSDAY	Thursday	
FRIDAY	Friday	
SATURDAY	Saturday	

DehumidificationOptions

Enumerator	Description	Notes
MECHANICAL_COOLING	Mechanical cooling	
DESICCANT	Desiccant	
SERIES_HEAT_RECOVERY	Series heat recovery	
NONE	None	
OTHER	Other	

DemandControlVentilationControlOptions

Enumerator	Description	Notes
CO2_RETURN_AIR	CO2 return air	
CO2_ZONE	CO2 zone	

Enumerator	Description	Notes
OTHER	Other	
NONE	None	

DrawPatternOptions

Enumerator	Description	Notes
VERY_SMALL	Very small	
LOW	Low	
MEDIUM	Medium	
HIGH	High	
OTHER	Other	

ElectricalPhaseOptions

Enumerator	Description	Notes
SINGLE_PHASE	Single Phase	
THREE_PHASE	Three Phase	

ElevatorOptions

Enumerator	Description	Notes
HYDRAULIC	Hydraulic	
TRACTION	Traction	
OTHER	Other	

EnergyRecoveryOperationOptions

Enumerator	Description	Notes
WHEN_FANS_ON	When fans on	
WHEN_MINIMUM_OUTSIDE_AIR	When minimum outside air	

Enumerator	Description	Notes
SCHEDULED	Scheduled	
OTHER	Other	
NONE	None	

EnergyRecoveryOptions

Enumerator	Description	Notes
SENSIBLE_HEAT_EXCHANGE	Sensible heat exchange	
ENTHALPY_HEAT_EXCHANGE	Enthalpy heat exchange	
SENSIBLE_HEAT_WHEEL	Sensible heat wheel	
ENTHALPY_HEAT_WHEEL	Enthalpy heat wheel	
HEAT_PIPE	Heat pipe	
OTHER	Other	
NONE	None	

EnergyRecoverySupplyAirTemperatureControlOptions

Enumerator	Description	Notes
FIXED_SETPOINT	Fixed setpoint	
MIXED_AIR_RESET	Mixed air reset	
OTHER	Other	
NONE	None	

EnergySourceOptions

Enumerator	Description	Notes
ELECTRICITY	Electricity	
NATURAL_GAS	Natural gas	

Enumerator	Description	Notes
PROPANE	Propane	
FUEL_OIL	Fuel oil	
STEAM	Steam	Only applicable for absorption chillers
NONE	None	No energy consumption by a utility supplied source occurs
OTHER	Other	

ExternalFluidSourceOptions

Enumerator	Description	Notes
CHILLED_WATER	Chilled water	
HOT_WATER	Hot water	
STEAM	Steam	

FanSpecificationMethodOptions

Enumerator	Description	Notes
SIMPLE	Simple	Specify the electric power input of fan
DETAILED	Detailed	Specify the brake horse power, design pressure rise through, total efficiency, motor efficiency

FanSystemOperationOptions

Enumerator	Description	Notes
CYCLING	Cycling	
CONTINUOUS	Continuous	
KEEP_OFF	Off	
OTHER	Other	

FanSystemSupplyFanControlOptions

Enumerator	Description	Notes
CONSTANT	Constant	
VARIABLE_SPEED_DRIVE	Variable speed drive	
MULTISPEED	Multispeed	
INLET_VANE	Inlet vane	
DISCHARGE_DAMPER	Discharge damper	
OTHER	Other	

FanSystemSupplyFanVolumeResetOptions

Enumerator	Description	Notes
CONSTANT	Constant	
DESIGN_LOAD_RESET	Design Load Reset	
OPERATING_CAPACITY_RESET	Operating Capacity Reset	
OTHER	Other	

FanSystemTemperatureControlOptions

Enumerator	Description	Notes
CONSTANT	Constant	
OUTDOOR_AIR_RESET	Outdoor air reset	
ZONE_RESET	Zone reset	
LOAD_RESET_TO_SPACE_TEMPERATURE	Load Reset To Space Temperature	
LOAD_RESET_DIFFERENTIAL_TEMPERATURE	Load Reset Differential Temperature	
SCHEDULED	Scheduled	
OTHER	Other	

FluidLoopFlowControlOptions

Enumerator	Description	Notes
<code>FIXED_FLOW</code>	Fixed flow	
<code>VARIABLE_FLOW</code>	Variable flow	

FluidLoopOperationOptions

Enumerator	Description	Notes
<code>CONTINUOUS</code>	Continuous	
<code>INTERMITTENT</code>	Intermittent/on-demand	
<code>SCHEDULED</code>	Scheduled	

FluidLoopOptions

Enumerator	Description	Notes
<code>HEATING</code>	Heating	
<code>COOLING</code>	Cooling	
<code>HEATING_AND_COOLING</code>	Heating and cooling	
<code>CONDENSER</code>	Condenser	
<code>OTHER</code>	Other	

HeatRejectionFanOptions

Enumerator	Description	Notes
<code>AXIAL</code>	Axial or Propellor	
<code>CENTRIFUGAL</code>	Centrifugal	
<code>OTHER</code>	Other	

HeatRejectionFanSpeedControlOptions

Enumerator	Description	Notes
CONSTANT	Constant	
TWO_SPEED	Two Speed	
VARIABLE_SPEED	Variable Speed	
OTHER	Other	

HeatRejectionFluidOptions

Enumerator	Description	Notes
WATER	Water	
REFRIGERANT	Refrigerant	Including R-448A
AMMONIA	Ammonia	
OTHER	Other	

HeatRejectionOptions

Enumerator	Description	Notes
OPEN_CIRCUIT_COOLING_TOWER	Open-circuit cooling tower	
CLOSED_CIRCUIT_COOLING_TOWER	Closed-circuit cooling tower or fluid cooler	
DRY_COOLER	Dry-cooler or air-cooled fluid cooler	
EVAPORATIVE_CONDENSER	Evaporative condenser	
AIR_COOLED_CONDENSER	Air cooled condenser	
OTHER	Other	

HeatingDesignDayOptions

Enumerator	Description	Notes
HEATING_99_6	Heating design day 99.6% annual cumulative frequency of occurrence	

Enumerator	Description	Notes
HEATING_99_0	Heating design day 99.0% annual cumulative frequency of occurrence	

HeatingMetricOptions

Enumerator	Description	Notes
HEAT_PUMP_COEFFICIENT_OF_PERFORMANCE_HIGH_TEMPERATURE	Efficiency at 8.3C/47F dry bulb and 6.1C/43F wet bulb	
HEAT_PUMP_COEFFICIENT_OF_PERFORMANCE_LOW_TEMPERATURE	Efficiency at -8.3C/17F dry bulb and -9.4C/15F wet bulb	
HEAT_PUMP_COEFFICIENT_OF_PERFORMANCE_HIGH_TEMPERATURE_NO_FAN	Efficiency not including fan at 8.3C/47F dry bulb and 6.1C/43F wet bulb	Used for heat pump and describes the efficiency not including the indoor supply fan. The outdoor evaporator fan power is generally included.
HEAT_PUMP_COEFFICIENT_OF_PERFORMANCE_LOW_TEMPERATURE_NO_FAN	Efficiency not including fan at -8.3C/17F dry bulb and -9.4C/15F wet bulb	Used for heat pump and describes the efficiency not including the indoor supply fan. The outdoor evaporator fan power is generally included.
THERMAL_EFFICIENCY	Et - thermal efficiency	
COMBUSTION_EFFICIENCY	Ec - combustion` efficiency	
ANNUAL_FUEL_UTILIZATION_EFFICIENCY	AFUE - annual fuel utilization efficiency	
HEATING_SEASONAL_PERFORMANCE_FACTOR	HSPF - heating seasonal performance factor	
HEATING_SEASONAL_PERFORMANCE_FACTOR_2	HSPF2 - heating seasonal performance factor 2	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_WATER_LOOP	Coefficient of performance at entering temperature of 20C/68F rated conditions for water loops per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_WATER_LOOP_NO_FAN	Coefficient of performance at entering temperature of 20C/68F rated conditions for water loops per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.

Enumerator	Description	Notes
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_GROUND_WATER	Coefficient of performance at entering temperature of 10C/50F rated conditions for ground water per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_AIR_GROUND_WATER_NO_FAN	Coefficient of performance not including fan at entering temperature of 10C/50F rated conditions for ground water per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_AIR_GROUND_LOOP	Coefficient of performance at entering temperature of 0C/32F rated conditions for ground water per ISO 13256-1	
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_AIR_GROUND_LOOP_NO_FAN	Coefficient of performance not including fan at entering temperature of 0C/32F rated conditions for ground water per ISO 13256-1	Used for heat pump and describes the efficiency not including the indoor supply fan.
COEFFICIENT_OF_PERFORMANCE_WATER_TO_WATER_WATER_LOOP	Coefficient of performance at entering temperature of 20C/68F rated conditions for water loops per ISO 13256-2	
COEFFICIENT_OF_PERFORMANCE_WATER_TO_WATER_GROUND_WATER	Coefficient of performance at entering temperature of 10C/50F rated conditions for ground water per ISO 13256-2	
COEFFICIENT_OF_PERFORMANCE_BRINE_TO_WATER_GROUND_LOOP	Coefficient of performance at entering temperature of 0C/32F rated conditions for ground water per ISO 13256-2	
NONE	None	
OTHER	Other	

HeatingSourceOptions

Enumerator	Description	Notes
ELECTRIC	Electric	

Enumerator	Description	Notes
HOT_WATER	Hot water	
NONE	None	
OTHER	Other	

HeatingSystemOptions

Enumerator	Description	Notes
HEAT_PUMP	Heat Pump	
FURNACE	Furnace	
ELECTRIC_RESISTANCE	Electric resistance	
FLUID_LOOP	Fluid loop	
NONE	None	
OTHER	Other	

HeatpumpAuxiliaryHeatOptions

Enumerator	Description	Notes
ELECTRIC_RESISTANCE	Electric resistance	
FURNACE	Furnace	
NONE	None	
OTHER	Other	

HumidificationOptions

Enumerator	Description	Notes
ADIABATIC	Adiabatic	
NONE	None	
OTHER	Other	

InfiltrationMethodOptions

Enumerator	Description	Notes
WEATHER_DRIVEN	Weather Driven. The amount of air leakage is determined by using the infiltration_flow_rate with a correlation usually involving windspeed, height, and the difference between indoor and outdoor temperature and is then multiplied by the schedule.	
PRESSURE_BASED	Pressure Based. The amount of air leakage is determined by induced airflows from pressure differences between zones, air distribution system components, the outside due to wind speed and direction.	
CONSTANT	Constant. The schedule is ignored.	
CONSTANT_SCHEDULED	Constant multiplied by the schedule.	
OTHER	Other infiltration methods.	

InsulationLocationOptions

Enumerator	Description	Notes
ABOVE_GROUND_WALL_EXTERIOR_CONTINUOUS	Continuous insulation on exterior of an above ground wall	
ABOVE_GROUND_WALL_INTERIOR_CONTINUOUS	Continuous insulation on interior of an above ground wall	
ABOVE_GROUND_WALL_FULL_CAVITY	Insulation within the cavity of an above ground wall that fills the entire cavity	
ABOVE_GROUND_WALL_PARTIAL_CAVITY	Insulation within the cavity of an above ground wall that fills only part of the cavity	
SLAB_HORIZONTAL_PERIMETER	Insulation on underside of the slab covering the perimeter	
SLAB_HORIZONTAL_FULL	Insulation fully covering the underside of the slab	
SLAB_VERTICAL	Insulation applied vertically next to edge of slab	
NONE	None	
OTHER	Other	

LightingDaylightingControlOptions

Enumerator	Description	Notes
STEPPED	Stepped	
CONTINUOUS_DIMMING	Continuous Dimming	
OTHER	Other types of daylighting control	
NONE	None	No daylighting is used.

LightingOccupancyControlOptions

Enumerator	Description	Notes
FULL_AUTO_ON	Full auto on	
PARTIAL_AUTO_ON	Parial auto on	
MANUAL_ON	Manual on	
OTHER	Other types of occupancy control	
NONE	None	No occupancy controls is used.

MiscellaneousEquipmentOptions

Enumerator	Description	Notes
PLUG	Plug	
PROCESS	Process	
INFORMATION_TECHNOLOGY_EQUIPMENT	Information technology equipment	
OTHER	Other	

PumpSpecificationMethodOptions

Enumerator	Description	Notes
SIMPLE	Simple	Specify the electric power input of pump
DETAILED	Detailed	Specify the motor nameplate power, design head, impellor efficiency, motor efficiency

PumpSpeedControlOptions

Enumerator	Description	Notes
FIXED_SPEED	Fixed speed	
TWO_SPEED	Two Speed	
VARIABLE_SPEED	Variable speed	
OTHER	Other	

RefrigeratedCaseEquipmentCategoryOptions

Enumerator	Description	Notes
HORIZONTAL_OPEN	Horizontal open	
HORIZONTAL_SOLID_DOOR	Horizontal solid door	
HORIZONTAL_TRANSPARENT_DOOR	Horizontal transparent door	
SEMIVERTICAL_OPEN	Semivertical open	
SERVICE_OVER_COUNTER	Service over counter	
VERTICAL_OPEN	Vertical open	
VERTICAL_SOLID_DOOR	Vertical solid door	
VERTICAL_TRANSPARENT_DOOR	Vertical transparent door	
OTHER	Other	

RefrigeratedCaseOptions

Enumerator	Description	Notes
COMMERCIAL_REFRIGERATION	Commercial refrigeration	
COMMERCIAL_REFRIGERATOR_SOLID_DOOR	Commercial refrigerator solid door	
COMMERCIAL_REFRIGERATOR_TRANSPARENT_DOOR	Commercial refrigerator transparent door	
COMMERCIAL_FREEZER_SOLID_DOOR	Commercial freezer solid door	
COMMERCIAL_FREEZER_TRANSPARENT_DOOR	Commercial freezer transparent door	

Enumerator	Description	Notes
COMMERCIAL_PULLDOWN_REFRIGERATOR	Commercial pulldown refrigerator	
COMMERCIAL_REFRIGERATOR_FREEZER_SOLID_DOOR	Commercial refrigerator freezer solid door	
OTHER	Other	

ScheduleOptions

Enumerator	Description	Notes
MULTIPLIER_DIMENSIONLESS	Multiplier dimensionless	
TEMPERATURE	Temperature	
POWER	Power	
FLOW_RATE	Flow rate	

ScheduleSequenceOptions

Enumerator	Description	Notes
HOURLY	Hourly	
EVENT	Event	

ServiceWaterHeaterTankOptions

Enumerator	Description	Notes
CONSUMER_INSTANTANEOUS	Consumer instantaneous	Uses UEF
COMMERCIAL_INSTANTANEOUS	Commercial instantaneous	Uses TE
CONSUMER_STORAGE	Consumer storage	Uses UEF
COMMERCIAL_STORAGE	Commercial storage	Uses TE and SBL
RESIDENTIAL_DUTY_COMMERCIAL_INSTANTANEOUS	Residential-Duty Commercial Instantaneous	Uses UEF
INDIRECT	Indirect	
BOILER	Boiler	

Enumerator	Description	Notes
COMMERCIAL_PACKAGED_BOILER	Commercial Packaged Boiler	
OTHER	Other	

ServiceWaterHeatingConfigurationOptions

Enumerator	Description	Notes
HERS_PARALLEL_PIPING	HERS parallel piping	
HERS_PIPE_INSULATION_ALL_LINES	HERS pipe insulation of all lines	
HERS_RECIRCULATION_DEMAND_CONTROL_OCCUPANCY_SENSOR	HERS recirculation demand control occupancy sensor	
HERS_RECIRCULATION_DEMAND_CONTROL_BUTTON	HERS recirculation demand control pull botton	
HERS_RECIRCULATION_NON_DEMAND_CONTROL	HERS recirculation non-demand control	
INSULATED_AND_PROTECTED_PIPE_BELOW_GRADE	Insulated and protected pipe below grade	
PARALLEL_PIPING	Parallel piping	
PIPE_INSULATION_ALL_LINES	Pipe insulation of all lines	
POINT_OF_USE	Point of use	
RECIRCULATION_DEMAND_CONTROL_OCCUPANCY_SENSOR	Recirculation demand control occupancy sensor	
RECIRCULATION_DEMAND_CONTROL_BUTTON	Recirculation demand control pull botton	
RECIRCULATION_NON_DEMAND_CONTROL	Recirculation non-demand control	
STANDARD	Standard	
OTHER	Other	

ServiceWaterHeatingEfficiencyMetricOptions

Enumerator	Description	Notes
ENERGY_FACTOR	Energy factor	

Enumerator	Description	Notes
UNIFORM_ENERGY_FACTOR	Uniform energy factor	See 10 CFR 430 Appendix E Uniform Test Method for Measuring the Energy Consumption of Water Heaters for more information.
THERMAL_EFFICIENCY	Thermal efficiency	
STANDBY_LOSS_FRACTION	Standby loss fraction	
STANDBY_LOSS_ENERGY	Standby loss fraction	Units are watts and can be converted from STANDBY_LOSS_FRACTION based formulat found in CBECC-Res User Manual
OTHER	Other	

ServiceWaterHeatingFixtureOptions

Enumerator	Description	Notes
SHOWER	Shower	
BATH	Bath	
RESTROOM_SINK	Restroom Sink	
DISHWASHER	Dishwasher	
KITCHEN_SINK	Kitchen sink	
WASH_SINK	Wash sink	
CLOTHES_WASHER	Clothes washing machine	
OTHER	Other	

ServiceWaterHeatingHeatRecoveryOptions

Enumerator	Description	Notes
NOT_APPLICABLE	Not applicable	
VERTICAL	Vertical	
HORIZONTAL	Horizontal	

Enumerator	Description	Notes
OTHER	Other	

ServiceWaterHeatingUseUnitOptions

Enumerator	Description	Notes
POWER_PER_PERSON	Power per person	The units for the use that correspond to this are watts/person
POWER_PER_AREA	Power per area	The units for the use that correspond to this are watts/m2
POWER	Power	The units for the use that correspond to this are watts
VOLUME_PER_PERSON	Volume per person	The units for the use of hot water that correspond to this are liters/(minute-person)
VOLUME_PER_AREA	Volume per area	The units for the use of hot water that correspond to this are liters/(minute-m2)
VOLUME	Volume	The units for the use of hot water that correspond to this are liters/minute
OTHER	Other	

SpaceFunctionOptions

Enumerator	Description	Notes
LABORATORY	Laboratory	
KITCHEN	Kitchen	
PLENUM	Plenum	A compartment or chamber, to which one or more ducts are connected, that forms a part of the air-distribution system and that is not used for occupancy or storage. The area of a plenum is not counted toward the floor area of a building.
CRAWL_SPACE	Crawl space	An unoccupied, unfinished, narrow space within a building, between the ground and the first (or ground) floor. The crawl space is so named because there is typically only enough room to crawl rather than stand; anything larger than about 1 to 1.5 metres (3 ft 3 in to 4 ft 11 in) and beneath the ground floor would tend to be considered a basement. The area of a crawl space is not counted toward the floor area of a building.

Enumerator	Description	Notes
INTERSTITIAL_SPACE	Interstitial space	The area of load bearing surfaces located above or below occupied building floors that is not available for general occupancy often due to inadequate clear headroom or lack of provisions for egress, and containing building structure or services predominantly serving adjacent floors or to provide access to such systems. They are not part of the air distribution system, conditioned, used for occupancy, or used for storage. Interstitial spaces often contain ducts, piping, conduit, and other components that serve the building systems. The area of a interstitial space is not counted toward the floor area of a building.
OTHER	Other	

StatusOptions

Enumerator	Description	Notes
NEW	New	
EXISTING	Existing	
EXISTING_PLUS_NEW	Existing plus new	
FUTURE	Future	
ALTERED	Altered	
ALTERED_ROOF_SURFACE	Altered roof surface	
OTHER	Other	

SubsurfaceClassificationOptions

Enumerator	Description	Notes
WINDOW	Window	
SKYLIGHT	Skylight	
DOOR	Door	
OTHER	Other types of subsurfaces that allow light to pass	

SubsurfaceDynamicGlazingOptions

Enumerator	Description	Notes
NOT_DYNAMIC	Not dynamic	
MANUAL_DYNAMIC	Manual dynamic	
AUTOMATIC_DYNAMIC	Automatic dynamic	

SurfaceAdjacencyOptions

Enumerator	Description	Notes
EXTERIOR	Exterior wall or roof which is adjacent to the exterior environment.	
GROUND	Slab-on-grad or below grade surface if adjacent to ground.	
INTERIOR	Interior surface if adjacent to another space which is explicitly modeled.	
IDENTICAL	Surface adjacent to a environment identical to the space. Sometimes this is described as adiabatic surface since no heat is transferred. The space on the other side of the surface is not explicitly modeled.	
UNDEFINED	The surface adjacency cannot be determined by the software.	

SurfaceClassificationOptions

Enumerator	Description	Notes
WALL	Vertical or nearly vertical wall	
FLOOR	Floor	
CEILING	Ceiling	

TemperatureResetOptions

Enumerator	Description	Notes
NO_RESET	No Reset	
OUTSIDE_AIR_RESET	Outside air reset	
LOAD_RESET	Load Reset	

Enumerator	Description	Notes
OTHER	Other	

TerminalFanConfigurationOptions

Enumerator	Description	Notes
PARALLEL	Parallel	
SERIES	Series	
OTHER	Other	

TerminalOptions

Enumerator	Description	Notes
VARIABLE_AIR_VOLUME	Variable air volume	
CONSTANT_AIR_VOLUME	Constant air volume	
RADIANT	Radiant	
BASEBOARD	Baseboard	
VARIABLE_REFRIGERANT_FLOW	Variable refrigerant flow indoor unit	
OTHER	Other	

TerminalTemperatureControlOptions

Enumerator	Description	Notes
CONSTANT	Constant	
LOAD_RESET_DIFFERENTIAL_TEMPERATURE	Load Reset Differential Temperature	
SCHEDULED	Scheduled	
OTHER	Other	

TransformerOptions

Enumerator	Description	Notes
DRY_TYPE	Dry Type	
FLUID_FILLED	Fluid Filled	
OTHER	Other	

WeatherFileDataSourceOptions

Enumerator	Description	Notes
HISTORIC_AGGREGATION	Historic data aggregated to represent typical weather	
HISTORIC_ACTUAL	Specific weather data for time period based on monitoring	
FUTURE	Weather data projected to represent future conditions	
OTHER	Other	

RulesetProjectDescription

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group.	ID			✓	All id data elements are shown as required so that that every data group can be explicitly identified. Almost no other data elements are shown as required but they may be required by specific rulesets.
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context

Name	Description	Data Type	Units	Range	Req	Notes
<code>ruleset_model_descriptions</code>	A list of the ruleset model descriptions within the project each with a unique <code>ruleset_model_type</code> .	<code>[[RuleSetModelDescription]]</code>			✓	For submissions with ruleset model descriptions (i.e., user, proposed, baseline,...) generated as separate files, only one model description would be defined for each ASHRAE229 ruleset project description json file. For submissions where the software always produces all ruleset model descriptions together, multiple model descriptions would appear in the same ASHRAE229 ruleset project description json file. Some rulesets may restrict the submission process including the number of files and number of model descriptions per file. Multiple model descriptions may be required by rulesets that require models at various tiers or levels of performance. Model descriptions may also be used by References between data groups and are restricted to be within the same ruleset model description. For 90.1-2019 appendix G model descriptions would be created for the following models: user, proposed, baseline at 0 degrees, baseline at 90 degrees, baseline at 180 degrees, and baseline at 270 degrees. All IDs within a ruleset model description should be unique for each type of data group. Between ruleset model descriptions, IDs should be the same for the same component. For example, a Surface with an ID of LobbyFrontSurface is very likely to have the same geometry for all ruleset model descriptions except perhaps the azimuth so that surface should have the ID LobbyFrontSurface for all ruleset model descriptions.
<code>metadata</code>	Metadata	<code>{Metadata}</code>				
<code>compliance_path</code>	Indicates the chosen compliance path if the ruleset has multiple compliance paths such as 90.1 Appendix G has code compliance and beyond code	<code><CompliancePathOptions2019ASHRAE901></code>				
<code>output_format_type</code>	Indicates the chosen format for output	<code>(<CommonOutputSchemaOptions>, <OutputSchemaOptions2019ASHRAE901>, <OutputSchemaOptions2019T24>, OutputSchemaOptionsRESNET)</code>				
<code>output</code>	Output	<code>({Output2019ASHRAE901})</code>				

Metadata

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Unique data set identifier	<code>ID</code>				Assigned by <i>data publisher</i> to identify this data. <code>id</code> shall remain unchanged for revisions of the same data.
<code>schema_author</code>	Name of the organization that published the schema	<code>String</code>			✓	Identifies the organization that defined the schema. Is should be set to: ASHRAE SPC 229 Schema Working Group
<code>schema_name</code>	Schema name or identifier	<code>String</code>			✓	Identifies the schema used to define the data content. It should be set to Ruleset Evaluation Schema
<code>schema_version</code>	The version of the schema the data complies with	<code>version</code>			✓	Common versions may include 0.0.36 from the first public review period. This is the value of the version field in the top of the ASRAE229.schema.json file or the version in the ASHRAE229.schema.yaml Schema data group.

Name	Description	Data Type	Units	Range	Req	Notes
<code>schema_url</code>	The Uniform Resource Locator (url) for the schema definition and/or documentation	String				Should be set to: https://github.com/open229/ruleset-model-description-schema
<code>author</code>	Name of the entity creating the serialization	String			✓	Identifies the organization that created the file
<code>description</code>	Description of data content (suitable for display)	String			✓	
<code>time_of_creation</code>	Timestamp indicating when the serialization was created	Timestamp			✓	Updated anytime any data content is modified.
<code>version</code>	Integer version identifier for the data	Version				Used by data publisher to track revisions of the data. Shall be incremented for each data revision
<code>source</code>	Source(s) of the data	String				Used by data publisher to document methods (e.g., software and version) used to generate data. Informative note: <code>source</code> may be different from other data source(s) included elsewhere within the data
<code>disclaimer</code>	Characterization of accuracy, limitations, and applicability of this data	String				
<code>notes</code>	Additional Information	String				

RulesetModelDescription

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	ID			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	String				
<code>notes</code>	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>type</code>	Describes the current ruleset model description for rulesets with multiple simulation models	<pre><CommonRulesetModelOptions>, <RulesetModelOptions2019ASHRAE90I>, <RulesetModelOptionsRESNET>, <RulesetModelOptions2019T24Com>, <RulesetModelOptions2019T24Res></pre>			✓	Will usually designate which simulation this RulesetModelDescription refers to such as: PROPOSED, BASELINE_0, USER, RATED, REFERENCE
<code>transformers</code>	Electrical transformers at the building site	[[Transformer]]				Contains a list of transformers that convert electricity from a higher voltage to one used by the building, exterior lighting, and other services at the site.
<code>buildings</code>	Buildings on the site	[[Building]]				Contains a list of buildings on the site (often just one).
<code>schedules</code>	Schedules for internal loads, thermostats, equipment operation and control, and any other need.	[[Schedule]]				Contains a list of schedules used in model.
<code>measured_infiltration_pressure_difference</code>	Differential pressure difference used during measurement for infiltration values.	Numeric	Pa	≥0		Used as rating conditions for air leakage for a building. The most common values used are 50 Pa or 75 Pa since they correspond to common rating conditions. For the model of actual building value this value would be measured but for created baseline or reference model this could be an assumed value.

Name	Description	Data Type	Units	Range	Req	Notes
is_measured_infiltration_based_on_test	Indicates whether the differential pressure difference used during measurement for infiltration values is based on pressure testing of the building.	Boolean				
altitude	Altitude of the building at ground level.	Numeric	m	>-1000, <10000		Altitude of the building site above sea level and used for adjusting airflows. If the altitude is not provided, it is assumed to be zero and airflows are already normalized to sea level.
fluid_loops	Fluid loops on the site	[[FluidLoop]]				Contains a list of fluid loops on the site that are not child loops. In primary/secondary piping systems or more complicated piping systems, this data element should include a list of the the primary loops. Child loops should appear in FluidLoop.child_loops.
service_water_heating_distribution_systems	Service water heating systems on the site	[[ServiceWaterHeatingDistributionSystem]]				Contains a list of service water heating distribution systems at the site.
service_water_heating_equipment	Service water heating equipment on the site	[[ServiceWaterHeatingEquipment]]				Contains a list of service water heating equipment at the site.
pumps	Pumps used on the site	[[Pump]]				
boilers	Boilers used on the site	[[Boiler]]				
chillers	Chillers used on the site	[[Chiller]]				
heat_rejections	HeatRejections used on the site	[[HeatRejection]]				
external_fluid_sources	ExternalFluidSources used on the site	[[ExternalFluidSource]]				The sources of energy, such as district chilled water, district hot water, and district steam.
site_zone_type	Site zone type for Sec 9.4.2	<ExteriorLightingZoneOptions2019ASHRAE901>				The site designation type for exterior lighting classification.
calendar	Information on the calendar used with the simulation.	{Calendar}				
weather	Information on the local weather conditions used with the simulation.	{Weather}				
model_output	Model output for a single instance	{(OutputInstance2019ASHRAE901)}				

Building

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
building_segments	Large portions of a building that share a building area type	[[BuildingSegment]]				Contains a list of building segments in the building.
elevators	Elevators	[[Elevator]]				Contains a list of elevators in the building.
exterior_lighting	Exterior lighting systems	[[ExteriorLighting]]				Contains a list of exterior lighting systems for the building.
refrigerated_cases	Refrigerated cases	[[RefrigeratedCase]]				Contains a list of refrigerated cases in the building.

Name	Description	Data Type	Units	Range	Req	Notes
<code>building_open_schedule</code>	ID of the Schedule indicating when the building is open	<code>Reference</code>				One represent when the building is open and zero when closed. If the schedule is not present, 24 hour operation is assumed. Constraint to use when implemented :Schedule:
<code>has_site_shading</code>	Indicates whether the site has features that cast shadows on the building	<code>Boolean</code>				
<code>number_of_floors_above_grade</code>	Number of floors above grade	<code>Numeric</code>		≥ 0		For the entire building including all building segments. Includes only floors that are not devoted solely to parking or unconditioned space.
<code>number_of_floors_below_grade</code>	Number of floors below grade	<code>Numeric</code>		≥ 0		For the entire building including all building segments. Includes only floors that are not devoted solely to parking or unconditioned space.

BuildingSegment

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>number_of_floors_above_grade</code>	Number of floors above grade	<code>Numeric</code>		≥ 0		For the portion of the building represented by the BuildingSegment.
<code>number_of_floors_below_grade</code>	Number of floors below grade	<code>Numeric</code>		≥ 0		For the portion of the building represented by the BuildingSegment.
<code>is_all_new</code>	Indicates whether the building segment is completely new construction (true) or existing (false).	<code>Boolean</code>				Projects that include additions should have a building segments that are existing (false) and for the addition (true). Curtain rules such as baseline fenestration area will apply differently to each portion.
<code>zones</code>	Zones in the building segment	<code>[[Zone]]</code>				Contains a list of zones in the building.
<code>heating_ventilating_air_conditioning_systems</code>	HVAC systems in the building segment	<code>[[HeatingVentilatingAirConditioningSystem]]</code>				Contains a list of HVAC systems in the building.
<code>area_type_vertical_fenestration</code>	Building area classification used for vertical fenestration	<code><verticalFenestrationBuildingAreaOptions2019ASHRAE90I></code>				The enumeration is based on the standard used.
<code>lighting_building_area_type</code>	Building area lighting area type	<code><LightingBuildingAreaOptions2019ASHRAE90I1T951TG38></code>				The lighting building area type for the building segment. If used, options are provided in the ruleset specific enumerations. This data element may also be used for determining the building area type for other purposes such as target setting targets (like 90.1 building performanc factors) so it should be completed even when the lighting building area method is not used.
<code>area_type_heating_ventilating_air_conditioning_system</code>	Classification used for HVAC	<code><heatingVentilatingAirConditioningBuildingAreaOptions2019ASHRAE90I></code>				The enumeration is based on the standard used.
<code>service_water_heating_area_type</code>	Service water heating area type classification	<code><ServiceWaterHeatingAreaOptions2019ASHRAE90I></code>				The enumeration is based on the standard used.
<code>service_water_heating_uses</code>	List of service water heating uses	<code>[[ServiceWaterHeatingUse]]</code>				This data element also appears in Space. A list of ServiceWaterHeatingUse data groups can either appear here or a list of references to IDs of a ServiceWaterHeatingUse already defined somewhere else or a mixture of both. A ServiceWaterHeatingUse that has one of the following unit types: [POWER_PER_PERSON, POWER_PER_AREA, VOLUME_PER_PERSON, VOLUME PER AREA] AND has not been explicitly referenced by any spaces is assumed to apply to each space in the building segment. A ServiceWaterHeatingUse that is referenced by one or more spaces will be applied only to those spaces. A ServiceWaterHeatingUse with units POWER, VOLUME, or OTHER will be applied only once to the entire building segment. When a reference is used, the constraint to use when implemented is to :ServiceWaterHeatingUse:

Zone

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	No multipliers or floor multipliers are used with the Zone data group so each zone should be individually identified.
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
spaces	Spaces in the zone	[[Space]]				Contains a list of spaces in the building.
floor_name	Floor name	String				Used to group zones on a floor. Zones with the same floor name are on the same floor. Numbers may be used as part of the floor name such as Level 1. If a number is used it should increase for increasing heights and use negative values for stories generally below ground. The floor name should generally correspond to numbering of floors on the plans.
volume	Volume of the space	Numeric	m3	≥0		
surfaces	Surfaces surrounding the zone	[[Surface]]				Contains a list of surfaces that define the zone.
conditioning_type	Space conditioning category	<ConditioningOptions>				
infiltration	Airleakage into the zone.	{Infiltration}				References a single infiltration data group. If a zone includes multiple spaces and infiltration is specified at the space level the rates and schedules should be aggregated to the zone.
design_thermostat_cooling_setpoint	Setpoint temperature for cooling during occupied hours	Numeric	C			The cooling setpoint temperature for the zone at design conditions.
thermostat_cooling_setpoint_schedule	ID of the Schedule containing the cooling setpoint temperatures	Reference				If the schedule is not present, the design thermostat cooling setpoint is assumed to be constant. Constraint to use when implemented :Schedule:
design_thermostat_heating_setpoint	Setpoint temperature for heating during occupied hours	Numeric	C			The heating setpoint temperature for the zone at design conditions.
thermostat_heating_setpoint_schedule	ID of the Schedule containing the heating setpoint temperatures	Reference				If the schedule is not present, the design thermostat heating setpoint is assumed to be constant. Constraint to use when implemented :Schedule:
minimum_humidity_setpoint_schedule	ID of the Schedule containing the minimum relative humidity setpoint	Reference		≥0, ≤1		For schedule values use relative humidity expressed on a 0 to 1 scale. Constraint to use when implemented :Schedule:
maximum_humidity_setpoint_schedule	ID of the Schedule containing the maximum relative humidity setpoint	Reference		≥0, ≤1		For schedule values use relative humidity expressed on a 0 to 1 scale. Constraint to use when implemented :Schedule:
terminals	List of terminals	[[Terminal]]				Multiple terminals may be used such as from a VAV system, a DOAS, and a baseboard.
served_by_service_water_heating_system	ID of the ServiceWaterHeatingDistributionSystem serving the zone	Reference				Contains the ID of the service water heating distribution system serving the zone. The ID is a unique identification for a specific ServiceWaterHeatingSystem data group. Constraint to use when implemented :ServiceWaterHeatingDistributionSystem:
transfer_airflow_rate	Airflow rate for transfer air	Numeric	L/s			Net transfer air. Positive values indicate transfer air in to the zone and negative values show transfer out of the zone.
transfer_airflow_source_zone	ID of the Zone for the source of transfer air.	Reference				Constraint to use when implemented :Zone:
zonal_supply_fans	A list of zonal supply fans	[[Fan]]				A list of references to one or more fans that is used to supply air to the zone. The fan should be configured to not be connected from any other HVAC data group. This may include fans used in a non-mechanical applications typically for warehouses with high air change rates.
supply_airflow_rate_multiplier_schedules	A list of IDs for Schedules containing the multipliers for the zonal supply airflow rates	Reference				Each schedule corresponds to the list in zonal_supply_fans. If a schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
zonal_exhaust_fans	A list of zonal exhaust fans	[[Fan]]				A list of references to one or more fans that is used to exhaust air from the zone. The fan should be configured to not be connected from any other HVAC data group. This may include fans used in a non-mechanical applications typically for warehouses with high air change rates.
exhaust_airflow_rate_multiplier_schedules	A list of IDs for Schedules containing the multipliers for the zonal exhaust airflow rates	Reference				Each schedule corresponds to the list in zonal_exhaust_fans. If a schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
makeup_airflow_rate	Airflow rate for makeup air	Numeric	L/s	≥0		
air_distribution_effectiveness	Air distribution effectiveness	Numeric		≥0		The zone air distribution effectiveness as defined in ASHRAE Standard 62.1 and shown as Ez.
aggregation_factor	Aggregation factor	Numeric		≥1		A factor that has already been applied when modeling a zone that represents an aggregation of several individual zones shown on design documents. For example, if five dwelling units facing the same orientation are modeled as a single zone, the total volume of the zone and the total surface area of the zone should be five times the values of a single dwelling unit shown on the design documents and aggregation_factor should be set to 5. Another example, when three identical floors are modeled, the zones modeled for these floors should represent the volume of the zone on three floors, and the aggregation_factor should be set to 3. A final example would be a big box retail store with eight units serving the core zone, the total floor area of the spaces in the core zone should already account for the total area, and the aggregation_factor should be set to 8. The factor is chiefly used to determine the size range for heating and cooling equipment serving that zone. Used only to confirm the individual size of heating or cooling units serving the zone so that the efficiency for that size range can be looked up in tables of equipment efficiency by size range. By dividing the heating and cooling capacity of the zone by the aggregation_factor, the result would indicate the capacity of the individual units represented by equipment. For zones not representing any type of aggregation, this value should be one. The aggregation_factor is typically an integer but occasionally may include a fractional value. The total area of the spaces and the volume of the zone should already be multiplied by the factor.

Space

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>interior_lighting</code>	Internal lighting that produce internal gains for a space.	<code>[{InteriorLighting}]</code>				References one or more InteriorLighting data groups that collectively define all interior lighting within the space.
<code>miscellaneous_equipment</code>	Miscellaneous equipment loads that produce internal gains for a space.	<code>[{MiscellaneousEquipment}]</code>				References one or more MiscellaneousEquipment that collectively describe all of the miscellaneous equipment in the space.
<code>floor_area</code>	The floor area of the space.	<code>Numeric</code>	m2	≥0		The floor area of a space within the building, including basements, mezzanine and intermediate-floored tiers, and penthouses with a headroom height of 7.5 ft or greater. It is measured from the exterior faces of walls or from the center-line of walls separating buildings, but excluding covered walkways, open roofed-over areas, porches and similar spaces, pipe trenches, exterior terraces or steps, chimneys, roof overhangs, and similar features. This is the floor area that is modeled.
<code>number_of_occupants</code>	Number of occupants in the space	<code>Numeric</code>		≥0		
<code>occupant_multiplier_schedule</code>	ID of the Schedule containing the multiplier for the number of occupants	<code>Reference</code>				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
<code>occupant_sensible_heat_gain</code>	Sensible heat gain of each occupant.	<code>Numeric</code>	W	≥0		
<code>occupant_latent_heat_gain</code>	Latent heat gain of each occupant.	<code>Numeric</code>	W	≥0		
<code>status_type</code>	Choice of new, existing, addition, alteration, etc.	<code><StatusOptions></code>				
<code>function</code>	Generic function for the space.	<code><SpaceFunctionOptions></code>				The enumeration is based on the standard used.
<code>envelope_space_type</code>	Envelope space type classification	<code><EnvelopeSpaceOptions2019ASHRAE901></code>				The enumeration is based on the standard used. Often referred to as occupancy type.
<code>lighting_space_type</code>	Lighting space type classification	<code><LightingSpaceOptions2019ASHRAE901TG37></code>				The enumeration is based on the standard used.
<code>ventilation_space_type</code>	Ventilation space type classification	<code><VentilationSpaceOptions2019ASHRAE901></code>				The enumeration is based on the standard used.
<code>service_water_heating_area_type</code>	Service water heating area type classification	<code><ServiceWaterHeatingAreaOptions2019ASHRAE901></code>				The enumeration is based on the standard used.
<code>service_water_heating_uses</code>	List of service water heating uses	<code>[({ServiceWaterHeatingUse}, Reference)]</code>				This data element also appears in BuildingSegment. A list of ServiceWaterHeatingUse data groups can either appear here or a list of references to IDs of a ServiceWaterHeatingUse already defined somewhere else or a mixture of both. References must refer to a ServiceWaterHeatingUse at the BuildingSegment level and that BuildingSegment must be the BuildingSegment to which this space belongs. ServiceWaterHeatingUses stored at the space level may only be used for this space and may not be referenced by other spaces. When a reference is used, the constraint to use when implemented is to :ServiceWaterHeatingUse:

Infiltration

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>modeling_method</code>	The software methodology chosen for modeling infiltration	<code><InfiltrationMethodOptions></code>				
<code>algorithm_name</code>	Name of the algorithm used for modeling infiltration in the specific simulation engine.	<code>String</code>				
<code>measured_air_leakage_rate</code>	Measured air leakage rate from infiltration of outside air	<code>Numeric</code>	L/s	≥0		Based on the pressure described in ASHRAE229.measured_infiltration_pressure_difference. If a zone includes multiple spaces and infiltration is specified at the space level the rates should be aggregated to the zone.
<code>flow_rate</code>	Design infiltration flow rate	<code>Numeric</code>	L/s	≥0		Infiltration flow rate for simulation infiltration models unadjusted for temperature difference or windspeed or schedule often with a windspeed at 10 mph (4.5 m/s). This may vary in meaning between simulation engines. If a zone includes multiple spaces and infiltration is specified at the space level the rates should be aggregated to the zone.
<code>multiplier_schedule</code>	ID of the Schedule containing the multiplier for the infiltration	<code>Reference</code>				If the schedule is not present, the multiplier is always one. If a zone includes multiple spaces and infiltration is specified at the space level the rates and schedules should be aggregated to the zone. Constraint to use when implemented :Schedule:

Surface

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>subsurfaces</code>	Subsurfaces that are on the surface	<code>[{Subsurface}]</code>				Contains a list of surfaces that define the space.
<code>classification</code>	Classification for the surface.	<code><SurfaceClassificationOptions></code>				Options for surface being interior or exterior wall, floor, or ceiling.
<code>area</code>	area of the surface	<code>Numeric</code>	m2	≥0		Measured from interior face area. It is the gross area of the wall and includes the area of all subsurfaces.
<code>tilt</code>	Angle between vertical and the surface outward normal	<code>Numeric</code>	degrees			Example value would be 0 = roof, 90 = wall, 180 = downward facing surface (exterior floor)

Name	Description	Data Type	Units	Range	Req	Notes
azimuth	Clockwise angle between North and the horizontal projection of the wall's outward normal.	Numeric	degrees	≥0		Example values would be 0 = north, 90 = East, 180 = South, 270 = West. For rulesets with baseline rotations, the angle of the azimuth should be different for each baseline rotation.
adjacent_to	Used to classify the conditions on the surface.	(<SurfaceAdjacencyOptions>, <AdditionalSurfaceAdjacencyOptionsRESNET>, <AdditionalSurfaceAdjacencyOptions2019ASHRAE901>)				Determines whether the other side of the surface is modeled and if not what assumptions should be used.
adjacent_zone	ID of the adjacent Zone for interior surface. Only required when adjacent zone is explicitly modeled when adjacent_to is set to INTERIOR.	Reference				Each interior surface between two zones should be described only once as a surface. No duplication of an interior surface facing each zone is necessary. Constraint to use when implemented :Zone:
does_cast_shade	Determines whether the surface is modeled as casting shade on other exterior surfaces	Boolean				
construction	Construction description of surface or an ID of another Construction.	({Construction}, Reference)				The construction data group can either appear here or a reference to an ID of a Construction used in a previous Surface. When the reference is used, the constraint to use when implemented is to :Construction:
optical_properties	Optical properties of the surface.	{SurfaceOpticalProperties}				
status_type	Choice of new, existing, addition, alteration, etc. for each ruleset.	<StatusOptions>				

Construction

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
classification	Classification for the subsurface being window, skylight, door.	(<CommonConstructionClassificationOptions>, <ConstructionClassificationOptions2019ASHRAE901>, <ConstructionClassificationOptions2019T24>)				
fraction_framing	Fraction of the construction that is framing.	Numeric		≥0, ≤1		Fraction of the construction using framing_layers, the remaining portion uses the primary_layers. If blank, assume zero framing.
primary_layers	List of names, or IDs, of layer descriptions starting from the outside surface for primary heat path	[({Material}, Reference)]				For constructions with framing and cavity heat transfer paths, use this for the cavity. For constructions with homogeneous layer, use this element only. Air films should not be included in the list of layers. When Construction.primary_layers are specified, the Construction.r_values should not be used. The list of primary_layers should include at least one Material that has mass with Material.density and Material.specific_heat specified. A list of material data groups can either appear here or a list of references to IDs of a Materials already defined somewhere else or a mixture of both. When a reference is used, the constraint to use when implemented is to :Material:

Name	Description	Data Type	Units	Range	Req	Notes
<code>framing_layers</code>	List of names, or IDs, of layer descriptions starting from the outside surface for the framing heat path	<code>[({Material}), Reference)]</code>				For constructions with framing and cavity heat transfer paths, use this for the framing otherwise leave blank. Air films should not be included in the list of layers. Only used when <code>primary_layers</code> is also populated. A list of material data groups can either appear here or a list of references to IDs of a Materials already defined somewhere else or a mixture of both. When a reference is used, the constraint to use when implemented is to <code>:Material</code> :
<code>insulation_locations</code>	List of locations for the insulation related to the surface that correspond to the values in the <code>r_values</code> list. Only used when <code>r_values</code> are specified.	<code>[<InsulationLocationOptions>]</code>				The items in the list of <code>r_values</code> should directly correspond to the items in the list of <code>insulation_locations</code> .
<code>u_factor</code>	surface U-factor	<code>Numeric</code>	W/m2-K	<code>≥0</code>		Includes interior and exterior air films as specified by the referenced standard.
<code>c_factor</code>	surface C-factor	<code>Numeric</code>	W/m2-K	<code>≥0</code>		Typically used to describe the thermal performance of constructions used in below grade walls. It is the time rate of steady-state heat flow through unit area of a material or construction, induced by a unit temperature difference between the body surfaces. Note that the C-factor does not include soil or air films.
<code>f_factor</code>	surface F-factor	<code>Numeric</code>	W/m-K	<code>≥0</code>		Typically used to describe the thermal performance of constructions used in slab-on-grade floors. It is the heat transfer through the floor, induced by a unit temperature difference between the outside and inside air temperature, on the per linear length of the exposed perimeter of the floor.
<code>r_values</code>	List of r-value of the insulation for the surface that correspond to values in the <code>insulation_locations</code> list	<code>[Numeric]</code>	K-m2/W	<code>≥0</code>		When <code>Construction.r_values</code> are specified, the <code>Construction.primary_layers</code> should not be used. The items in the list of <code>r_values</code> should directly correspond to the items in the list of <code>insulation_locations</code> .
<code>has_radiant_heating</code>	Includes embedded radiant heating elements	<code>Boolean</code>				
<code>has_radiant_cooling</code>	Includes embedded radiant cooling elements	<code>Boolean</code>				

Material

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if <code>id</code> is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>thickness</code>	The thickness of the material layer	<code>Numeric</code>	m	<code>>0</code>		
<code>thermal_conductivity</code>	The thermal conductivity of the material layer	<code>Numeric</code>	W/m-K	<code>≥0</code>		When <code>thermal_conductivity</code> is specified, <code>r_value</code> should not be provided.

Name	Description	Data Type	Units	Range	Req	Notes
density	The density of the material layer	Numeric	kg/m3	≥0		
specific_heat	The specific heat of the material layer	Numeric	J/kg-K	≥0		
r_value	r-value of the insulation for the material layer	Numeric	K-m2/W	≥0		When r_value is specified, thermal_conductivity should not be provided. Typically used for insulation or air gaps.

SurfaceOpticalProperties

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
absorptance_thermal_exterior	Thermal absorptance of long wavelength radiation on the exterior surface.	Numeric		≥0		May also be called thermal emittance, emittance or emissivity and represents the fraction of incident long wavelength radiation that is absorbed by the material
absorptance_solar_exterior	Thermal absorptance of short wavelength radiation on the exterior surface.	Numeric		≥0		Equals one minus the solar reflectance (for opaque materials) and represents the fraction of incident solar radiation that is absorbed by the material
absorptance_visible_exterior	Thermal absorptance of visible radiation on the exterior surface.	Numeric		≥0		Equals one minus the visible reflectance (for opaque materials) and represents the fraction of incident visible wavelength radiation that is absorbed by the material

Name	Description	Data Type	Units	Range	Req	Notes
<code>absorptance_thermal_interior</code>	Thermal absorptance of long wavelength radiation on the interior surface.	Numeric		≥ 0		May also be called thermal emittance, emittance or emissivity and represents the fraction of incident long wavelength radiation that is absorbed by the material
<code>absorptance_solar_interior</code>	Thermal absorptance of short wavelength radiation on the interior surface.	Numeric		≥ 0		Equals one minus the solar reflectance (for opaque materials) and represents the fraction of incident solar radiation that is absorbed by the material
<code>absorptance_visible_interior</code>	Thermal absorptance of visible radiation on the interior surface.	Numeric		≥ 0		Equals one minus the visible reflectance (for opaque materials) and represents the fraction of incident visible wavelength radiation that is absorbed by the material

Subsurface

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	ID			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	String				
<code>notes</code>	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>classification</code>	Classification for the subsurface being window, skylight, door.	<SubsurfaceClassificationOptions>				
<code>subclassification</code>	Standard specific subclassification for subsurfaces	<SubsurfaceSubclassificationOptions2019ASHRAE901>				
<code>is_operable</code>	Identifies whether window subsurface can be opened and closed including by pivoting or sliding.	Boolean				This applies to windows and skylights but not to doors.
<code>has_open_sensor</code>	Has sensor and reports to building control system when the window or door is open.	Boolean				
<code>framing_type</code>	The material of the framing.	<SubsurfaceFrameOptions2019ASHRAE901>				This applies to windows and skylights but not to doors.
<code>glazed_area</code>	Area of subsurface including glass and transparent surfaces	Numeric	m2	≥ 0		
<code>opaque_area</code>	Area of subsurface framing for a window or skylight or opaque portion for a door.	Numeric	m2	≥ 0		
<code>u_factor</code>	Overall Subsurface U-factor	Numeric	W/m2-K	≥ 0		Includes interior and exterior air films as specified by the referenced standard.

Name	Description	Data Type	Units	Range	Req	Notes
<code>dynamic_glazing_type</code>	Type of dynamic glazing for the window subsurface	<code><SubsurfaceDynamicGlazingOptions></code>				Indicates if the glazed subsurface can change it's performance properties and if it is automatic or not.
<code>solar_heat_gain_coefficient</code>	Subsurface SHGC	<code>Numeric</code>		≥ 0		For dynamic glazing represents the minimum SHGC
<code>maximum_solar_heat_gain_coefficient</code>	Maximum Subsurface SHGC for Dynamic Glazing	<code>Numeric</code>		≥ 0		Only used for dynamic glazing
<code>visible_transmittance</code>	Subsurface VT	<code>Numeric</code>		≥ 0		For dynamic glazing represents the maximum visible transmittance
<code>minimum_visible_transmittance</code>	Minimum Subsurface VT for Dynamic Glazing	<code>Numeric</code>		≥ 0		Only used for dynamic glazing
<code>depth_of_overhang</code>	Distance from the edge of the overhang to the subsurface.	<code>Numeric</code>	m	≥ 0		
<code>has_shading_overhang</code>	Identifies whether subsurface has overhangs	<code>Boolean</code>				
<code>has_shading_sidefins</code>	Identifies whether subsurface has sidefins	<code>Boolean</code>				
<code>has_manual_interior_shades</code>	Are there manually-operated interior shading such as blinds, curtains or shades	<code>Boolean</code>				
<code>solar_transmittance_multiplier_summer</code>	Solar transmittance multiplier for summer	<code>Numeric</code>		≥ 0		Often used to account for interior shading such as drapes.
<code>solar_transmittance_multiplier_winter</code>	Solar transmittance multiplier for summer	<code>Numeric</code>		≥ 0		Often used to account for interior shading such as drapes.
<code>has_automatic_shades</code>	Are there automatic interior shading such as blinds, curtains or shades	<code>Boolean</code>				
<code>status_type</code>	Choice of new, existing, addition, alteration, etc. for each ruleset.	<code><StatusOptions></code>				

InteriorLighting

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>purpose_type</code>	Lighting purpose type classification	<code><LightingPurposeOptions2019ASHRAE901></code>				The enumeration is based on the standard used.
<code>power_per_area</code>	Total power for lights divided by the area of the space.	<code>Numeric</code>	W/m2			When computing the power per area use the area of the entire space.
<code>lighting_multiplier_schedule</code>	ID of the Schedule containing the multiplier for lighting	<code>Reference</code>				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
<code>occupancy_control_type</code>	Indicates the type of occupancy controls	<code><LightingOccupancyControlOptions></code>				
<code>daylighting_control_type</code>	Indicates the type of daylighting controls	<code><LightingDaylightingControlOptions></code>				
<code>are_schedules_used_for_modeling_occupancy_control</code>	Indicates that schedule values are used for modeling the impacts of occupancy controls on lighting.	<code>Boolean</code>				
<code>are_schedules_used_for_modeling_daylighting_control</code>	Indicates that schedule values are used for modeling the impacts of daylighting controls on lighting.	<code>Boolean</code>				For simulations that are modeling daylighting by computing the illuminance this should be false.

MiscellaneousEquipment

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
energy_type	Source of energy for the miscellaneous equipment in the space	<EnergySourceOptions>				To indicate that the energy is not accounted for by any utility supplied source, choose NONE. To indicate equipment with a energy source from a Loop choose OTHER
power	Power for miscellaneous equipment in the space	Numeric	W			The value of power for the miscellaneous equipment in the space that when multiplied by the schedule is the consumption of power (for electricity or other energy_type's) for each period in the schedule.
multiplier_schedule	ID of the Schedule containing the multiplier for miscellaneous equipment power in the space.	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
sensible_fraction	Fraction of energy that is a sensible load on the space.	Numeric		≥0, ≤1		Sensible plus latent do not necessarily add up to 1.0.
latent_fraction	Fraction of energy that is a latent load on the space.	Numeric		≥0, ≤1		Sensible plus latent do not necessarily add up to 1.0.
remaining_fraction_to_loop	ID of the FluidLoop for the remaining fraction	Reference				If sensible_fraction and latent_fraction add up to less than 1.0 and this field is specified then the chilled water or condenser water loop specified gets the remaining fraction of the miscellenous load. Constraint to use when implemented :FluidLoop:
energy_from_loop	ID of the FluidLoop serving as the source of energy	Reference				References the loop that is the source of energy for the miscellenous equipment. Example would be hot water that is used for laundry in the zone. When this is used the energy_type should be set to OTHER. Constraint to use when implemented :FluidLoop:
type	Type of miscellaneous equipment	<MiscellaneousEquipmentOptions>				
automatic_controlled_percentage	Fraction of receptacles that have automatic controls.	Numeric		≥0, ≤1		

Transformer

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	

Name	Description	Data Type	Units	Range	Req	Notes
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	The type of transformer	<TransformerOptions>				
phase	The number of electrical phases	<ElectricalPhaseOptions>				
efficiency	Transformer efficiency	Numeric		≥0, ≤1		Expresses the efficiency of the transformer as a fraction from 0 to 1, where 1 would represent 100% efficiency.
capacity	Rated Capacity of the Transformer	Numeric	V-A	≥0		
peak_load	Annual Peak electric load on the transformer	Numeric	W	≥0		Peak electric load on the transformer based on an annual simulation with typical weather file.

Schedule

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
sequence_type	Schedule sequence type	<ScheduleSequenceOptions>				
hourly_values	Hourly Values of Schedule	[Numeric] [0..8784]				Used when schedule_sequence_type is HOURLY. Can also use functions like EFLH(), MAX(), MIN() to determine overall characteristics for the list of schedule values. A year that does not have a leap day should be a list of 8760 values and a leap year should be a list of 8784 values. All schedules for a project should be consistently either a list of 8760 values or a list of 8784 values but not mixed within a project. The list of values is in standard time and should be adjusted when Calendar.has_daylight_saving_time is set to True based on local rules governing the start and end dates of daylighting savings time.
hourly_heating_design_day	Hourly values for heating design day	[Numeric] [0..24]				Used when schedule_sequence_type is HOURLY. Populate only one of the data elements: hourly_heating_design_day and hourly_heating_design_year, not both.

Name	Description	Data Type	Units	Range	Req	Notes
hourly_heating_design_year	Hourly values for heating design year	[Numeric/Null][0..8784]				Used when schedule_sequence_type is HOURLY. Use when more than a 24 hour period was used for sizing. Values with the value Null will be ignored. Populate only one of the data elements: hourly_heating_design_day and hourly_heating_design_year, not both. A year that does not have a leap day should be a list of 8760 values and a leap year should be a list of 8784 values. All schedules for a project should be consistently either a list of 8760 values or a list of 8784 values but not mixed within a project. The list of values is in standard time and should be adjusted when Calendar.has_daylight_saving_time is set to True based on local rules governing the start and end dates of daylighting savings time.
hourly_cooling_design_day	Hourly values for cooling design day	[Numeric][0..24]				Used when schedule_sequence_type is HOURLY. Populate only one of the data elements: hourly_cooling_design_day and hourly_cooling_design_year, not both.
hourly_cooling_design_year	Hourly values for cooling design year	[Numeric/Null][0..8784]				Used when schedule_sequence_type is HOURLY. Use when more than a 24 hour period was used for sizing. Values with the value Null will be ignored. Populate only one of the data elements: hourly_cooling_design_day and hourly_cooling_design_year, not both. A year that does not have a leap day should be a list of 8760 values and a leap year should be a list of 8784 values. All schedules for a project should be consistently either a list of 8760 values or a list of 8784 values but not mixed within a project. The list of values is in standard time and should be adjusted when Calendar.has_daylight_saving_time is set to True based on local rules governing the start and end dates of daylighting savings time.
event_times	Event times when the schedule changes	[Numeric]	s			Used when schedule_sequence_type is EVENT to describe the time of the year in seconds that the schedule changes value.
event_values	Event value at corresponding event time.	[Numeric]				Used when schedule_sequence_type is EVENT. New values starting at corresponding to the event time until following event time minus one second. Can also use functions like EFLH(), MAX(), MIN() to determine overall characteristics for the list of schedule values.
event_times_heating_design_day	Event times when the schedule changes for heating design day	[Numeric]	s			Used when schedule_sequence_type is EVENT.
event_values_heating_design_day	Event value at corresponding event time for heating design day	[Numeric]				Used when schedule_sequence_type is EVENT. New values starting at corresponding to the event time until following event time minus one second.
event_times_cooling_design_day	Event times when the schedule changes for cooling design day	[Numeric]	s			Used when schedule_sequence_type is EVENT.
event_values_cooling_design_day	Event value at corresponding event time for cooling design day	[Numeric]				Used when schedule_sequence_type is EVENT. New values starting at corresponding to the event time until following event time minus one second.
type	The type of schedule	<ScheduleOptions>				Primarily indicates if the values may be represented by units such as C for temperature or W for power or m3/s for flow rate or are dimensionless multipliers.
prescribed_type	Options when any schedule values have changed from what appears in the schedule library	<PrescribedScheduleOptions2019ASHRAE90I>				Some rule sets prescribe which schedules to use for modeling purposes. This option indicates if the schedule is as prescribed or has been modified.
is_modified_for_workaround	True if any schedule has been modified for a workaround	Boolean				Not all technologies can be modeled directly in the simulation program. Modifying schedules is a common workaround for modeling some technologies. This flag is used to indicate when a schedule has been modified for this purpose.

Calendar

Name	Description	Data Type	Units	Range	Req	Notes
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context

Name	Description	Data Type	Units	Range	Req	Notes
day_of_week_for_january_1	Day of the week for January 1	<DayOfWeekOptions>				
has_daylight_saving_time	The schedules adjust for Daylight Saving Time	Boolean				This value should be true if the modeled calendar includes daylight saving time. Schedule.hourly_values are entered in standard time and should be adjusted if this value is true. The adjustment period should be based on local rules governing the start and end dates of daylighting savings time.

Weather

Name	Description	Data Type	Units	Range	Req	Notes
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
ground_temperature_schedule	ID of the Schedule containing ground temperatures	Reference				Constraint to use when implemented :Schedule:
file_name	The file name for the weather file including extension.	String				The file name for the annual weather file such as from TMY, TRY, CWEC, CTZ, WYEC or other sources.
data_source_type	Data source use for the weather file.	<WeatherFileDataSourceOptions>				
climate_zone	The designation of the climate zone where the building is located	<ClimateZoneOptions2019ASHRAE90I>				The enumeration is based on the standard used.
cooling_dry_bulb_design_day_type	The frequency of occurrence type for cooling design day based on dry-bulb temperature	<CoolingDesignDayOptions>				
evaporation_wet_bulb_design_day_type	The frequency of occurrence type for evaporation wet-bulb design day	<CoolingDesignDayOptions>				This may also be known as cooling design wet-bulb temperature day type.
heating_dry_bulb_design_day_type	The frequency of occurrence type for heating design day based on dry-bulb temperature	<HeatingDesignDayOptions>				

Elevator

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	The type of elevator	<ElevatorOptions>				

Name	Description	Data Type	Units	Range	Req	Notes
number_of_floors_served	Number of floors served by the elevator. This includes all unique full height building levels adjacent to the elevator shaft that are between the top and bottom floors where the elevator opens. Exclude levels serving only parking garages or other levels explicitly not part of the building.	Numeric		≥1		
motor_location_zone	ID of the Zone where the heat from the motor goes	Reference				When specified, it is the zone the heat from the elevator motor get added to. Constraint to use when implemented :Zone:
motor_heat_fraction	Fraction of heat from the motor that is added as a sensible load zone.	Numeric		≥0, ≤1		
cab_location_zone	ID of the Zone where the heat from the cab goes	Reference				When specified, it is the zone the heat from the cab get added to to the zone including lighting heat, fan heat, and accessory heat. Constraint to use when implemented :Zone:
cab_heat_fraction	Fraction of heat from the cab that is added as a sensible load zone.	Numeric		≥0, ≤1		When specified, it is the fraction of the heat from the cab that gets added to the zone including lighting heat, fan heat, and accessory heat
is_variable_speed_motor	If the elevator uses a variable speed motor drive	Boolean				
motor_power	Elevator average hourly peak motor power	Numeric	W			The motor power can be provided either together with or, instead of, the detailed elements used to calculate it.
cab_counterweight	Elevator car counterweight	Numeric	kg			
cab_weight	Weight of elevator car	Numeric	kg			
design_load	Elevator load at which to operate	Numeric	kg			
speed	Design speed of the elevator	Numeric	m/s			
cab_area	Floor area of elevator cab	Numeric	m2			
cab_lighting_power	Lighting power of cab	Numeric	W			
cab_ventilation_fan	Ventilation fan for the cab	{Fan}				
cab_ventilation_fan_multiplier_schedule	ID of the Schedule containing the multiplier for elevator ventilation fan operation	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
cab_motor_multiplier_schedule	ID of the Schedule containing the multiplier for the elevator motor operation	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
cab_lighting_multiplier_schedule	ID of the Schedule containing the multiplier for the elevator lighting	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:

HeatingVentilatingAirConditioningSystem

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
fan_system	Fan system	{FanSystem}				One FanSystem for each HeatingVentilatingAirConditioningSystem so if a direct outdoor air system is used a second Zone Terminal should be specified with a separate HeatingVentilatingAirConditioningSystem.
heating_system	Heating system	{HeatingSystem}				References the HeatingSystem data group that acts as the main heating system or coil for this HVAC System. If the HVAC system does not have a heating system, this entry can be left blank, or the HeatingSystem HeatingSystemOptions can be set to equal NONE.
cooling_system	Cooling system	{CoolingSystem}				References the CoolingSystem data group associated with this HVAC System. If the HVAC system does not have a cooling system, this entry can be left blank, or the CoolingSystem CoolingSystemOptions can be set to equal NONE.
preheat_system	Pre-heating system	{HeatingSystem}				References a HeatingSystem data group that provides preheat to this HVAC System. Preheat is typically a coil used to temper cold air entering an HVAC systm often when high quantities of outdoor air is expected. If the HVAC system does not have a preheat system, this entry can be left blank, or the HeatingSystem HeatingSystemOptions can be set to equal NONE in the referenced data group.
humidification_type	Humidification type	<HumidificationOptions>				
status_type	Choice of new, existing, addition, alteration, etc. for each ruleset.	<StatusOptions>				

HeatingSystem

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	Heating system type	<HeatingSystemOptions>				System configurations that typically are at the zone and include a compressor (such as packaged terminal air conditioning, packaged terminal heat pumps, window air conditioning units, and water loop heat pumps) should be reported in the schema using HeatingSystem and CoolingSystem. Systems that include gas or electric furnaces should be reported in the schema using HeatingSystem. Distributed systems where each zone is individually served by dedicated fans and/or coils (such as four-pipe fan coil, two-pipe fan coil, radiant systems, baseboards, chilled beams, and variable refrigerant flow indoor units) should be reported in the schema using Terminal with the cooling and heating described in the cooling_source and heating_source data elements (and any other relevant Terminal Data elements). Evaporative cooling systems should be described in CoolingSystem. Passive diffusers with no coil or fan should be described in Terminal.
energy_source_type	Source of energy for the heating system	<EnergySourceOptions>				
hot_water_loop	ID of the FluidLoop providing the hot water	Reference				Constraint to use when implemented :FluidLoop:
water_source_heat_pump_loop	ID of the FluidLoop containing water source heat pumps	Reference				Constraint to use when implemented :FluidLoop:

Name	Description	Data Type	Units	Range	Req	Notes
design_capacity	Design heating capacity	Numeric	W	≥0		Design capacity may be determined by sizing performed by the software. This value should be provided if sized by the simulation software as an output or explicitly provided as a simulation input.
rated_capacity	Rated heating capacity	Numeric	W	≥0		At rating conditions. The rated capacity is generally used to determine efficiency level when efficiency level is based on size. It is net heating capacity taking into account the heat from the indoor fan.
oversizing_factor	The oversizing factor applied to the peak load that results in the heat capacity. Zero indicates no oversizing. Negative numbers indicate undersizing.	Numeric				Used for furnace or heat pump.
is_calculated_size	True if the component is sized by software	Boolean				Determined in accordance with generally accepted engineering standards. For capacities determined by software, including the simulation software, this should be true. For capacities that are autosized values should be true. Typically when true, the sizing is based on one or more design days, or other time periods, and should be consistent with the weather.heating_design_day_type. For capacities reflecting the an actual piece of equipment this should be false.
heating_coil_setpoint	Setpoint of the air leaving the heating coil	Numeric	C			The setpoint of the air leaving the heating coil at design conditions.
efficiency_metric_values	List of efficiency metric values that correspond to the descriptors in efficiency_metric_types list	[Numeric]	W/W			Used for furnace or heat pump.
efficiency_metric_types	List of efficiency metrics that correspond to the values in efficiency_metric_values list	[<HeatingMetricoptions>]				Used for furnace or heat pump.
heatpump_auxiliary_heat_type	Heatpump auxiliary heat type used for backup	<HeatpumpAuxiliaryHeatoptions>				
heatpump_auxiliary_heat_high_shutoff_temperature	Heatpump auxiliary heat high temperature shutoff	Numeric	C			The auxiliary heat is available at or below this outside temperature to satisfy the load. If this temperature is higher than heatpump_low_shutoff_temperature then between the two temperatures, the heat pump is energized first and if any load remains, then the auxiliary heat is energized.
heatpump_low_shutoff_temperature	Heatpump low temperature shutoff	Numeric/Null	C			The heat pump is available at above this outdoor temperature to satisfy the load. The heat pump is always energized first to meet any heating load prior to the use of auxiliary heat. For heat pumps with no lower temperature limit, this value should not be provided or else set to Null.

CoolingSystem

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	Cooling system type	<CoolingSystemOptions>				System configurations that typically are at the zone and include a compressor (such as packaged terminal air conditioning, packaged terminal heat pumps, window air conditioning units, and water loop heat pumps) should be reported in the schema using HeatingSystem and CoolingSystem. Systems that include gas or electric furnaces should be reported in the schema using HeatingSystem. Distributed systems where each zone is individually served by dedicated fans and/or coils (such as four-pipe fan coil, two-pipe fan coil, radiant systems, baseboards, chilled beams, and variable refrigerant flow indoor units) should be reported in the schema using Terminal with the cooling and heating described in the cooling_source and heating_source data elements (and any other relevant Terminal Data elements). Evaporative cooling systems should be described in CoolingSystem. Passive diffusers with no coil or fan should be described in Terminal.
design_total_cool_capacity	Design total cooling capacity	Numeric	W	≥0		Designed total cooling capacity. Design capacity may be determined by sizing performed by the software. This value should be provided if sized by the simulation software as an output or explicitly provided as a simulation input.
design_sensible_cool_capacity	Design sensible cooling capacity	Numeric	W	≥0		Designed sensible cooling capacity. Design sensible capacity may be determined by sizing performed by the software. This value should be provided if sized by the simulation as an output or explicitly provided as a simulation input.

Name	Description	Data Type	Units	Range	Req	Notes
<code>rated_total_cool_capacity</code>	Rated total cooling capacity	<code>Numeric</code>	W	≥ 0		At rating conditions. The rated capacity is generally used to determine efficiency level when efficiency level is based on size. It is net cooling capacity taking into account the heat from the indoor fan.
<code>rated_sensible_cool_capacity</code>	Rated sensible cooling capacity	<code>Numeric</code>	W	≥ 0		At rating conditions. The rated capacity is generally used to determine efficiency level when efficiency level is based on size. It is net sensible cooling capacity taking into account the heat from the indoor fan.
<code>oversizing_factor</code>	The oversizing factor applied to the peak load that results in the cool capacity. Zero indicates no oversizing. Negative numbers indicate undersizing.	<code>Numeric</code>				
<code>is_calculated_size</code>	True if the component is sized by software	<code>Boolean</code>				Determined in accordance with generally accepted engineering standards. For capacities determined by software, including the simulation software, this should be true. For capacities that are autosized values should be true. Typically when true, the sizing is based on one or more design days, or other time periods, and should be consistent with the <code>weather.cooling_design_day_type</code> . For capacities reflecting the an actual piece of equipment this should be false.
<code>chilled_water_loop</code>	ID of the FluidLoop providing chilled water	<code>Reference</code>				Constraint to use when implemented :FluidLoop:
<code>condenser_water_loop</code>	ID of the FluidLoop providing condenser water	<code>Reference</code>				Constraint to use when implemented :FluidLoop:
<code>efficiency_metric_values</code>	List of efficiency metric values that correspond to the descriptors in <code>efficiency_metric_types</code> list	<code>[Numeric]</code>	W/W			Used for direct expansion.
<code>efficiency_metric_types</code>	List of efficiency metrics that correspond to the values in <code>efficiency_metric_values</code> list	<code>[<CoolingMetricoptions>]</code>				Used for direct expansion.
<code>dehumidification_type</code>	Dehumidification type	<code><DehumidificationOptions></code>				
<code>cooling_turndown_ratio</code>	Cooling turndown ratio	<code>Numeric</code>				Cooling capacity turndown before simultanenous heating and cooling occurs.

FanSystem

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>supply_fans</code>	List of supply fans	<code>[{Fan}]</code>				List of fans that supply air by the HVAC system and are located in the building.
<code>return_fans</code>	List of return fans	<code>[{Fan}]</code>				List of fans that return air to the HVAC system and are located in the building.
<code>exhaust_fans</code>	List of exhaust fans	<code>[{Fan}]</code>				List of fans that exhaust air from the building either at the air handler or located in various spaces. The fans are located in the building.
<code>relief_fans</code>	List of relief fans	<code>[{Fan}]</code>				List of fans that exhaust air from the building at the air handler to maintain proper pressurization. The fans are located in the building.
<code>air_economizer</code>	Air side economizer related to the fan system	<code>{AirEconomizer}</code>				
<code>air_energy_recovery</code>	Air side energy recovery related to the fan system	<code>{AirEnergyRecovery}</code>				
<code>temperature_control</code>	Supply air temperature control type	<code><FanSystemTemperatureControlOptions></code>				
<code>operation_during_occupied</code>	Operation during occupied times type	<code><FanSystemOperationOptions></code>				
<code>operation_during_unoccupied</code>	Operation during unoccupied times type	<code><FanSystemOperationOptions></code>				

Name	Description	Data Type	Units	Range	Req	Notes
has_lock_out_central_heat_during_unoccupied	Locks out the use of central heating during unoccupied hours	Boolean				Typically used when zone has heating coil available that can serve unoccupied heating load.
fan_control	Supply fan control type	<FanSystemSupplyFanControlOptions>				
reset_differential_temperature	Supply air temperature reset differential temperature at minimum cooling load	Numeric	K			When temperature_control is LOAD_RESET_TO_SPACE_TEMPERATURE this temperate is added to the supply air temperature at minimum cooling load conditions. When temperature_control is LOAD_RESET_DIFFERENTIAL_TEMPERATURE this temperate is the temperate below space tempature when no cooling load. The supply air temperature is specified in Terminal.
supply_air_temperature_reset_load_fraction	Supply air temperature reset load fraction	Numeric				When temperature_control is LOAD_RESET_TO_SPACE_TEMPERATURE, this is the threshold fraction below which supply air temperature reset begins. Between the stated load fraction and 0% load the supply air temperature is ramped linearly until the supply air temperature reaches the zone. The supply air temperature is specified in Terminal.
supply_air_temperature_reset_schedule	ID of the Schedule for the supply air temperature reset	Reference				The supply air temperature is specified in Terminal. Constraint to use when implemented :Schedule:
fan_volume_reset_type	Fan volume reset control type	<FanSystemSupplyFanVolumeResetOptions>				
fan_volume_reset_fraction	Fan volume reset load fraction	Numeric				When fan_volume_reset_type is DESIGN_LOAD_RESET this is the fraction of the design load that corresponds to minimum air flow. When fan_volume_reset_type is OPERATING_CAPACITY_RESET this is the fraction of the instantaneous operating capacity that corresponds to minimum air flow.
operating_schedule	ID of the Schedule indicating operation	Reference				If the schedule is not present, the fansystem is always operating. Schedule values of zero when fan is scheduled to be off and one when the fan is scheduled to be on. While a zero indicates the fan is off, it may still be activated during that hour by other controls such as providing fan cycling at night to maintain the unoccupied set point. No values beside 0 and 1 are allowed. Constraint to use when implemented :Schedule:
minimum_airflow	Minimum volume airflow	Numeric	L/s			The minimum airflow may be determined by sizing performed by the softare. This value should be provided if sized by the simulation software as an output or explicitly provided as a simulation input.
minimum_outdoor_airflow	Minimum outdoor air volume airflow	Numeric	L/s			If the minimum_outdoor_airflow is equal to the maximum_outdoor_airflow and equal to the total supply fan volume that indicates a constant volume dedicated outdoor air system. If only the maximum_outdoor_airflow and equal to the total supply fan volume but the minimum_outdoor_airflow is lower that indicates an air side economizer.
maximum_outdoor_airflow	Maximum outdoor air volume airflow	Numeric	L/s			If the minimum_outdoor_airflow is equal to the maximum_outdoor_airflow and equal to the total supply fan volume that indicates a constant volume dedicated outdoor air system. If only the maximum_outdoor_airflow and equal to the total supply fan volume but the minimum_outdoor_airflow is lower that indicates an air side economizer.
air_filter_merv_rating	The MERV rating of the air filter	Numeric		≥1, ≤20		
has_fully_ducted_return	If the fan system has fully ducted return.	Boolean				
demand_control_ventilation_control	Demand control ventilation control type	<DemandControlVentilationControlOptions>				

AirEconomizer

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	Type	<AirEconomizerOptions>				
high_limit_shutoff_temperature	High limit temperature shutoff	Numeric	C			

Name	Description	Data Type	Units	Range	Req	Notes
<code>is_integrated</code>	True if the economizer and mechanical cooling system can work together	Boolean				When the economizer is integrated with the mechanical cooling system it is capable of providing partial cooling even when additional mechanical cooling is required to meet the remainder of the cooling load.

AirEnergyRecovery

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	ID			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	String				
<code>notes</code>	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>type</code>	Energy recovery type	<EnergyRecoveryOptions>				
<code>enthalpy_recovery_ratio</code>	Enthalpy recovery ratio	Numeric				The change in the enthalpy of the outdoor air supply divided by the difference between the outdoor air and entering exhaust air enthalpy, expressed as a percentage.
<code>energy_recovery_operation</code>	Energy recovery operation	<EnergyRecoveryOperationOptions>				
<code>energy_recovery_supply_air_temperature_control</code>	Energy recovery supply air temperature control	<EnergyRecoverySupplyAirTemperatureControlOptions>				
<code>design_sensible_effectiveness</code>	Design sensible effectiveness	Numeric				
<code>design_latent_effectiveness</code>	Design latent effectiveness	Numeric				
<code>outdoor_airflow</code>	Outdoor airflow	Numeric	L/s			
<code>exhaust_airflow</code>	Exhaust airflow	Numeric	L/s			

Fan

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	ID			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	String				
<code>notes</code>	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>design_airflow</code>	Design airflow	Numeric	L/s			The design airflow may be determined by sizing performed by the software. This value should be provided if sized by the simulation software as an output or explicitly provided as a simulation input.

Name	Description	Data Type	Units	Range	Req	Notes
is_airflow_calculated	True if the airflow is sized by software	Boolean				Determined in accordance with generally accepted engineering standards. For air flows determined by software, including the simulation software, this should be true. For air flows that are autosized values should be true. Typically when true, the sizing is based on one or more design days, or other time periods, and should be consistent with the weather.heating_design_day_type and weather.cooling_design_day_type. For air flows reflecting the an actual piece of equipment this should be false. If the air flow is a calculated sum of hardcoded zone air flow values this should be false.
specification_method	Options for how the fan is specified	<FanSpecificationMethodOptions>				Indicates if the fan is specified using the SIMPLE method of just the design electric power or using the DETAILED method of using pressure rise, nameplate power, shaft power, total efficiency, and motor efficiency.
design_electric_power	Design electric fan power	Numeric	W			Only used when specification_method is set to Simple.
design_pressure_rise	Pressure rise through fan at design flow conditions	Numeric	Pa			Only used when specification_method is set to Detailed
motor_nameplate_power	nameplate power of fan motor	Numeric	W			Only used when specification_method is set to Detailed.
shaft_power	fan shaft power	Numeric	W			Power delivered to the fans shaft and does not include the mechanical drive losses. Equivalent to fan brake horsepower for inch-pound units. Only used when specification_method is set to Detailed.
total_efficiency	Total fan efficiency	Numeric		≥0, ≤1		Only used when specification_method is set to Detailed. It includes the overall efficiency of the combination of the fan, drive, and motor.
motor_efficiency	Fan motor efficiency	Numeric		≥0, ≤1		Only used when specification_method is set to Detailed.
motor_heat_to_airflow_fraction	Fraction of motor heat added to the airflow.	Numeric		≥0, ≤1		Fraction to airflow plus fraction to zone do not necessarily add up to 1.0.
motor_heat_to_zone_fraction	Fraction of motor heat added to the zone.	Numeric		≥0, ≤1		Fraction to airflow plus fraction to zone do not necessarily add up to 1.0.
motor_location_zone	ID of the Zone where the heat from the motor goes	Reference				When specified, it is the zone the heat from the fan motor get added to. Constraint to use when implemented :Zone:
status_type	Choice of new, existing, addition, alteration, etc. for each ruleset.	<StatusOptions>				
operating_points	Operating points	[{FanOperatingPoint}]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The airflow and power should be directly related. A minimum number of four points is recommended. Certain rulesets may have a different minimum number of points. For example, 90.1 Appendix G expects 11 data points. The ruleset checking specification would normally provide a list operating points that should match what is provided by the simulation.

FanOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
airflow	Airflow	Numeric	L/s			No name and id is needed since typically used as one of a series.
power	power	Numeric	W			The rate of energy used by the fan at the given airflow.

Terminal

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>type</code>	Type of terminal	<code><TerminalOptions></code>				System configurations that typically are at the zone and include a compressor (such as packaged terminal air conditioning, packaged terminal heat pumps, window air conditioning units, and water loop heat pumps) should be reported in the schema using <code>HeatingSystem</code> and <code>CoolingSystem</code> . Systems that include gas or electric furnaces should be reported in the schema using <code>HeatingSystem</code> . Distributed systems where each zone is individually served by dedicated fans and/or coils (such as four-pipe fan coil, two-pipe fan coil, radiant systems, baseboards, chilled beams, and variable refrigerant flow indoor units) should be reported in the schema using <code>Terminal</code> with the cooling and heating described in the <code>cooling_source</code> and <code>heating_source</code> data elements (and any other relevant <code>Terminal</code> Data elements). Evaporative cooling systems should be described in <code>CoolingSystem</code> . Passive diffusers with no coil or fan should be described in <code>Terminal</code> .
<code>served_by_heating_ventilating_air_conditioning_system</code>	ID of the <code>HeatingVentilatingAirConditioningSystem</code> serving the terminal	<code>Reference</code>				Contains ID of the HVAC system serving the terminal - from Unique Identification Number in <code>HeatingVentilatingAirConditioningSystem</code> . Constraint to use when implemented : <code>HeatingVentilatingAirConditioningSystem</code> :
<code>heating_source</code>	Source of heating	<code><HeatingSourceOptions></code>				Used for terminal heating including reheat.
<code>heating_from_loop</code>	ID of the <code>FluidLoop</code> providing heating	<code>Reference</code>				Only used when <code>heating_source</code> is hot water. Used for terminal heating including reheat. Constraint to use when implemented : <code>FluidLoop</code> :
<code>cooling_source</code>	Source of cooling	<code><CoolingSourceOptions></code>				Used for terminal cooling.
<code>cooling_from_loop</code>	ID of the <code>FluidLoop</code> providing cooling	<code>Reference</code>				Only used when <code>cooling_source</code> is chilled water. Used for terminal cooling including radiant and fan coils. Constraint to use when implemented : <code>FluidLoop</code> :
<code>fan</code>	Terminal fan	<code>{Fan}</code>				The fan data group associated with the terminal. If no fan is modeled for this terminal, this field should be left blank.
<code>fan_configuration</code>	Fan configuration	<code><TerminalFanConfigurationOptions></code>				
<code>primary_airflow</code>	Zone terminal primary airflow	<code>Numeric</code>	L/s			Zone terminal primary airflow at design conditions.
<code>secondary_airflow</code>	Zone terminal secondary airflow	<code>Numeric</code>	L/s			
<code>maximum_heating_airflow</code>	Zone terminal maximum heating airflow	<code>Numeric</code>	L/s			Entry only needed if maximum heating airflow is different than the pimary airflow such as when describing a dual maximum VAV box control.
<code>supply_design_heating_setpoint_temperature</code>	Zone terminal supply design heating temperature setpoint	<code>Numeric</code>	C			
<code>supply_design_cooling_setpoint_temperature</code>	Zone terminal supply design cooling temperature setpoint	<code>Numeric</code>	C			
<code>temperature_control</code>	Temperature control type	<code><TerminalTemperatureControlOptions></code>				
<code>minimum_airflow</code>	Zone terminal minimum volume airflow	<code>Numeric</code>	L/s			
<code>minimum_outdoor_airflow</code>	Zone terminal minimum outdoor air volume airflow	<code>Numeric</code>	L/s			This value should be provided if fan airflow determined by sizing performed by the software or explicitly provided. This should be the design flow of outdoor air based on an occupied space using the appropriate ventilation standard such as ASHRAE Standard 62.1. Adjustments to lower the value for demand controlled ventilation should be made using the <code>minimum_outdoor_airflow_multiplier_schedule</code> .
<code>minimum_outdoor_airflow_multiplier_schedule</code>	ID of the <code>Schedule</code> containing the multiplier for zone terminal minimum outdoor air volume airflow	<code>Reference</code>				If the schedule is not present, the multiplier is always one. Often used to indicate the reduced flow needed when the space has reduced occupancy when a demand controlled ventilation control is used. Constraint to use when implemented : <code>Schedule</code> :
<code>heating_capacity</code>	Heating capacity for baseboard or radiant system or reheat	<code>Numeric</code>	W			Only includes the heating capacity of the terminal for hot water or electric coil. This value should be provided if determined by sizing performed by the software or explicitly provided.
<code>cooling_capacity</code>	Cooling capacity for the radiant system or cooling coil	<code>Numeric</code>	W			Only includes the cooling capacity of the terminal for chilled water coil for radiant or fan coil. This value should be provided if determined by sizing performed by the software or explicitly provided.
<code>is_supply_ducted</code>	True if the the supply is ducted.	<code>Boolean</code>				Packaged terminal air conditioners (PTAC), packaged terminal heat pumps (PTHP), and window or through the wall air conditioners should be <code>False</code> and other DX systems including rooftop units should be <code>True</code>
<code>has_demand_control_ventilation</code>	True if the zone has demand control ventilation	<code>Boolean</code>				The zone is either served by an air handler that responds to demand control ventilation signals from that zone or is served by a DOAS that specifically provides air to that zone on demand.
<code>is_fan_first_stage_heat</code>	True if the the only source of first stage heating is the heat from the fan itself	<code>Boolean</code>				Only appropriate with variable volume boxes that are fan powered

FluidLoop

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>type</code>	Type of loop	<code><FluidLoopOptions></code>				
<code>pump_power_per_flow_rate</code>	Total design pump power divided by the loop design flow rate	<code>Numeric</code>		W-s/L		This is the pump power per flow rate for the entire pumping system on the current <code>FluidLoop</code> . The power and flow rate should be for the current <code>FluidLoop</code> only and does not include power and flow rate in any child loops.

Name	Description	Data Type	Units	Range	Req	Notes
child_loops	Other fluid loops connected to this one as children.	[[FluidLoop]]				Secondary loops should be described as child loops.
cooling_or_condensing_design_and_control	Fluid loop design and control used for cooling or condensing loops	{FluidLoopDesignAndControl}				References a FluidLoopDesignAndControl data group which describes how this FluidLoop is sized and controlled in cooling or condensing mode.
heating_design_and_control	Fluid loop design and control used for heating loops	{FluidLoopDesignAndControl}				References a FluidLoopDesignAndControl data group which describes how this FluidLoop is sized and controlled in heating mode.

FluidLoopDesignAndControl

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
design_supply_temperature	Design Supply Temperature	Numeric	C			
design_return_temperature	Design Return Temperature	Numeric	C			
is_sized_using_coincident_load	True if the loop is sized based on coincident load	Boolean				
minimum_flow_fraction	Minimum fraction of full flow allowed	Numeric				The minimum flow rate expressed as a fraction of the loop design flow rate.
operation	Type of operation used by loop	<FluidLoopOperationOptions>				
operation_schedule	ID of the Schedule describing operation	Reference				One represents when the fluid loop is available to be operating and zero when not available to be operating. Only used when operation equals SCHEDULED. Constraint to use when implemented :Schedule:
flow_control	Flow control options	<FluidLoopFlowControlOptions>				
temperature_reset_type	Type of temperature reset used by loop	<TemperatureResetOptions>				
outdoor_high_for_loop_supply_reset_temperature	Outdoor high for loop supply temp reset	Numeric	C			Used when temperature_reset_type = OUTSIDE_AIR_RESET
outdoor_low_for_loop_supply_reset_temperature	Outdoor low for loop supply temp reset	Numeric	C			Used when temperature_reset_type = OUTSIDE_AIR_RESET
loop_supply_temperature_at_outdoor_high	Loop supply temperature at outdoor high temperature	Numeric	C			Used when temperature_reset_type = OUTSIDE_AIR_RESET
loop_supply_temperature_at_outdoor_low	Loop supply temperature at outdoor low temperature	Numeric	C			Used when temperature_reset_type = OUTSIDE_AIR_RESET
loop_supply_temperature_at_low_load	Loop supply temperature at low load	Numeric	C			Used when temperature_reset_type = LOAD_RESET
has_integrated_waterside_economizer	True if chilled water loop described has an integrated waterside economizer	Boolean				

Pump

Name	Description	Data Type	Units	Range	Req	Notes
<code>id</code>	Scope-unique reference identifier for instances of this data group	<code>ID</code>			✓	
<code>reporting_name</code>	Descriptive name used in RCT reports if id is not already a descriptive name	<code>String</code>				
<code>notes</code>	Supplementary information to provide context to the model reviewer	<code>String</code>				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
<code>loop_or_piping</code>	ID of the FluidLoop or ServiceWaterPiping connected to the Pump	<code>Reference</code>			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :FluidLoop; or :ServiceWaterPiping;
<code>specification_method</code>	Options for how the pump is specified	<code><PumpSpecificationMethodOptions></code>				
<code>design_electric_power</code>	Pump design electric power	<code>Numeric</code>	W			Pump electric power at design conditions. Only used when <code>specification_method</code> is set to Simple
<code>motor_nameplate_power</code>	Pump motor nameplate power	<code>Numeric</code>	W			Only used when <code>specification_method</code> is set to Detailed
<code>design_head</code>	Head of the pump at design flow conditions	<code>Numeric</code>	Pa			Only used when <code>specification_method</code> is set to Detailed
<code>impeller_efficiency</code>	Full load efficiency of the impeller	<code>Numeric</code>		≥ 0 , ≤ 1		Only used when <code>specification_method</code> is set to Detailed
<code>motor_efficiency</code>	Full load efficiency of the pump motor	<code>Numeric</code>		≥ 0 , ≤ 1		Only used when <code>specification_method</code> is set to Detailed
<code>speed_control</code>	Options for pump speed control	<code><PumpSpeedControlOptions></code>				
<code>design_flow</code>	Design Pump Flowrate	<code>Numeric</code>	L/s			This value should be provided if determined by sizing performed by the software or explicitly provided.
<code>is_flow_calculated</code>	True if the <code>design_flow</code> is sized by software	<code>Boolean</code>				Determined in accordance with generally accepted engineering standards. For flows determined by software, including the simulation software, this should be true. For flows that are autosized values should be true. Typically when true, the sizing is based on one or more design days, or other time periods, and should be consistent with the <code>weather.heating_design_day_type</code> and <code>weather.cooling_design_day_type</code> . For flows reflecting the an actual piece of equipment this should be false.
<code>operating_points</code>	Operating points	<code>[{PumpOperatingPoint}]</code>				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The flow rate and power should be directly related. A minimum number of four points is recommended. The ruleset checking specification would normally provide a list operating points that should match what is provided by the simulation.

PumpOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
<code>flow</code>	Flow rate	<code>Numeric</code>	L/s			No name and id is needed since typically used as one of a series.

Name	Description	Data Type	Units	Range	Req	Notes
power	Power	Numeric	W			The rate of energy used per unit time by the pump at the given flow rate.

Boiler

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
loop	ID of the FluidLoop connected to the Boiler	Reference			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :FluidLoop:
design_capacity	Heating capacity	Numeric	W			
rated_capacity	Heating capacity	Numeric	W			At rating conditions.
minimum_load_ratio	Minimum fraction of full load allowed	Numeric				
draft_type	Combustion option	<BoilerCombustionOptions>				
energy_source_type	Source of energy for the boiler	<EnergySourceOptions>				
efficiency_metric_types	List of efficiency metrics that correspond to the values in efficiency_metric_values list	[<BoilerEfficiencyMetricoptions>]				
efficiency_metric_values	List of efficiency metric values that correspond to the descriptors in efficiency_metric_types list	[Numeric]		≥0		
operating_points	Operating points	[{BoilerOperatingPoint}]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The energy consumption per unit time should be directly related to the load on the boiler and entering water temperature. A minimum number of sixteen points is recommended. The ruleset checking specification would normally provide a list list operating points that should match what is provided by the simulation.
auxiliary_power	Auxiliary power	Numeric	W			Power for boiler pump, combustion fan, or other auxiliary that operates when boiler operates.
operation_lower_limit	Heating load range operation, lower limit	Numeric	W			The heating rate below which the boiler will not operate. This is often used in plants that have multiple boilers to specify how they should be sequenced.

Name	Description	Data Type	Units	Range	Req	Notes
operation_upper_limit	Heating load range operation, upper limit	Numeric	W			The heating rate above which the boiler will not operate. This is often used in plants that have multiple boilers to specify how they should be sequenced.

BoilerOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
load	Load	Numeric	W			No name and id is needed since typically used as one of a series.
entering_water_temperature	Entering water temperature	Numeric	C			Temperature of water entering the boiler which returned from the loop. This entry is optional since not all curves for boiler performance need this.
power	Power	Numeric	W			Energy consumption per unit time.

Chiller

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
cooling_loop	ID of the FluidLoop connected to the Chiller for supplying chilled water	Reference			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :FluidLoop:
condensing_loop	ID of the FluidLoop connected to the Chiller that provides condensing water	Reference				No condensing loop name implies air-cooled chiller. Constraint to use when implemented :FluidLoop:
compressor_type	Compressor Type	<ChillerCompressorOptions>				
chiller_function	Function such as chilled and/or hot water	<ChillerFunctionOptions>				
energy_source_type	Source of energy for the chiller	<EnergySourceOptions>				
design_capacity	Chiller Design Cooling Capacity	Numeric	W			The capacity of the chiller at design conditions. This should correspond with the design_entering_condenser_temperature and design_leaving_evaporator_temperature.
rated_capacity	Chiller Rated Cooling Capacity	Numeric	W			The capacity of the chiller at rating conditions. This should correspond with the rated_entering_condenser_temperature and rated_leaving_evaporator_temperature.
rated_entering_condenser_temperature	Rated entering condenser temperature	Numeric	C			The rated outside air dry-bulb temperature for air cooled chillers or design entering condenser water temperature for water cooled chillers. For AHRI conditions for inch-pound ratings for air cooled chillers are 35C/95F and water cooled are 29.4C/85F. The AHRI conditions for SI ratings are 35C for air cooled chillers and 30C for water cooled. This should correspond with the rated capacity and the FULL_LOAD_EFFICIENCY_RATED and INTEGRATED_PART_LOAD_VALUE.
rated_leaving_evaporator_temperature	Rated leaving evaporator temperature	Numeric	C			For AHRI conditions for inch-pound ratings are 6.7C/44F. For AHRI conditions for SI ratings are 7C. This should correspond with the rated capacity and the FULL_LOAD_EFFICIENCY_RATED and INTEGRATED_PART_LOAD_VALUE.
minimum_unload_ratio	Minimum unload ratio of full load allowed	Numeric				The ratio of current load to available capacity at a given simulation time step. Used only when this value is higher than the minimum_load_ratio. Between the minimum_unload_ratio and the minimum_load_ratio, the chiller creates a false load in order to continue operating at the minimum_unload_ratio.
minimum_load_ratio	Minimum fraction of full load allowed	Numeric				The ratio of current load to available capacity at a given simulation time step. Below the minimum load ratio, the chiller cycles on and off.
design_flow_evaporator	Chiller evaporator design flow	Numeric	L/s			

Name	Description	Data Type	Units	Range	Req	Notes
design_flow_condenser	Chiller condenser design flow	Numeric	L/s			
design_entering_condenser_temperature	Design entering condenser temperature	Numeric	C			The temperature entering the condenser portion of the chiller at design conditions. The design outside air dry-bulb temperature for air cooled chillers or design entering condenser water temperature for water cooled chillers. This should correspond with the design_capacity and the FULL_LOAD_EFFICIENCY_DESIGN and NONSTANDARD_PART_LOAD_VALUE.
design_leaving_evaporator_temperature	Design leaving evaporator temperature	Numeric	C			The temperature leaving the evaporator portion of the chiller at design conditions. This should correspond with the design_capacity and the FULL_LOAD_EFFICIENCY_DESIGN and NONSTANDARD_PART_LOAD_VALUE.
efficiency_metric_values	List of efficiency metric values that correspond to the descriptors in efficiency_metric_types list	[Numeric]	W/W	≥0		The useful cooling energy supplied by the chiller divided by the electric energy into the chiller both in the same units when the chiller is supplying a predefined load.
efficiency_metric_types	List of efficiency metrics that correspond to the values in efficiency_metric_values list	[<ChillerEfficiencyMetricOptions>]				
capacity_operating_points	Capacity points	[{ChillerCapacityOperatingPoint}]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The capacity should be directly related to the chilled water supply temperature and condenser temperature. A minimum number of sixteen points is recommended. The ruleset checking specification would normally provide a list list operating points that should match what is provided by the simulation.
power_operating_points	Energy validation points	[{ChillerPowerOperatingPoint}]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The power should be directly related to the chilled water supply temperature, condenser temperature, and load on the chiller. An indicator that the chiller is at full capacity is also important to differentiate between fully loaded and partially loaded operation. A minimum number of 64 points is recommended but fewer may be adequate to confirm proper operation especially if the temperatures and load are not repeated. The ruleset checking specification would normally provide a list list operating points that should match what is provided by the simulation.
is_chilled_water_pump_interlocked	Indicates if the operation of the chilled water pump is interlocked with the operation of the chiller	Boolean				Should be set to true when the chiller and the chiller water pump are controlled to turn on an off at the same time.
is_condenser_water_pump_interlocked	Indicates if the operation of the condenser water pump is interlocked with the operation of the chiller	Boolean				Should be set to true when the chiller and the condenser water pump are controlled to turn on an off at the same time.
heat_recovery_loop	ID of the FluidLoop connected to the chiller for supplying recovered heat	Reference				Constraint to use when implemented :FluidLoop:
heat_recovery_fraction	Heat recovery fraction	Numeric		≥0, ≤1		Fraction of total rejected heat that can be recovered at full load

ChillerCapacityOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
chilled_water_supply_temperature	Chilled water supply temperature	Numeric	C			No name and id is needed since used as one of a series. The temperature is leaving the chiller.
condenser_temperature	Condenser temperature	Numeric	C			Outside air dry-bulb temperature for air cooled chillers and condenser water temperature for water cooled chillers. For water cooled chillers, this is the temperature as the water enters the chiller. For air cooled chillers this the temperature of the ambient air.
capacity	Capacity	Numeric	W			The capacity available by the chiller at the given operating point.

ChillerPowerOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
chilled_water_supply_temperature	Chilled water supply temperature	Numeric	C			No name and id is needed since used as one of a series. The temperature is leaving the chiller.
condenser_temperature	Condenser temperature	Numeric	C			Outside air dry-bulb temperature for air cooled chillers and condenser water temperature for water cooled chillers. For water cooled chillers, this is the temperature as the water enters the chiller. For air cooled chillers this the temperature of the ambient air.
load	Load	Numeric	W			
is_at_full_capacity	True if chiller is at full capacity	Boolean	W			True if chiller is operating at full copacity for the given temperature conditions
power	Power	Numeric	W			The rate of energy used by the chiller at the given operating point.

HeatRejection

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
loop	ID of the FluidLoop connected to the HeatRejection	Reference			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :FluidLoop:
type	Heat Rejection Type	<HeatRejectionOptions>				
fan_type	Heat Rejection Fan Type	<HeatRejectionFanOptions>				
fan	Heat Rejection Fan	{Fan}				
fluid	Fluid Cooled by Heat Rejection	<HeatRejectionFluidOptions>				
number_of_cells	Number of cells	Numeric		≥1		
range	Heat rejection Range	Numeric	K			
approach	Heat rejection Approach	Numeric	K			
fan_speed_control	Fan Speed Control Type	<HeatRejectionFanSpeedControlOptions>				

Name	Description	Data Type	Units	Range	Req	Notes
design_wetbulb_temperature	Design wetbulb temperature	Numeric	C			0.4% ASHRAE MCWB
design_water_flowrate	Design condenser water flow rate	Numeric	L/s			
rated_water_flowrate	Rated condenser water flow rate	Numeric	L/s			At rating conditions.
leaving_water_setpoint_temperature	leaving water setpoint temperature	Numeric	C			

ExternalFluidSource

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
loop	ID of the FluidLoop connected to the ExternalFluidSource	Reference			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :FluidLoop:
type	Type of external fluid source	<ExternalFluidSourceOptions>				
energy_source_type	Source of energy for the external fluid source	<EnergySourceOptions>				

ServiceWaterHeatingDistributionSystem

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
design_supply_temperature	Design supply temperature setpoint of service water heating loop	Numeric	C			From CBECC-Cor.
design_supply_temperature_difference	Design supply temperature difference (deltaT) of service water heating loop	Numeric	C			From CBECC-Cor.
tanks	Tanks within service water heating distribution system	[{Tank}]				Contains a list of storage tanks that are part of this service water heating distribution system but not part of individual service water heaters.
is_central_system	Indicates whether it is a central service water heater distribution system	Boolean				From CBECC-Cor.
service_water_piping	Connected service water piping	{ServiceWaterPiping}				Contains the service water piping that is associated with this service water heater distribution system.

Name	Description	Data Type	Units	Range	Req	Notes
distribution_compactness	Type of compact distribution system	<ServiceWaterHeatingDistributionCompactnessOptions2019T24Com>				From CBECC-Com.
control_type	Type of distribution system	<ServiceWaterHeatingControlOptions2019T24Com>				From CBECC-Com.
configuration_type	Type of configuration	<ServiceWaterHeatingConfigurationOptions>				From CBECC-Com.
is_recovered_heat_from_drain_used_by_water_heater	Indicates whether the recovered heat from the shower drain used by the service water heater	Boolean				From CBECC-Res.
drain_heat_recovery_efficiency	Shower heat drain recovery efficiency	Numeric		≥0, ≤1		From CBECC-Com. May use the Canadian Standards Association Rated Recovery Efficiency.
drain_heat_recovery_type	Drain heat recovery type	<ServiceWaterHeatingHeatRecoveryOptions>				From CBECC-Res.
flow_multiplier_schedule	ID of the Schedule containing a multiplier for the service water heating loop flow	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
entering_water_mains_temperature_schedule	ID of the Schedule containing the temperature for unheated entering water to the building site often referenced as mains temperature	Reference				Constraint to use when implemented ::
is_ground_temperature_used_for_entering_water	Indicates whether ground temperature is the source of the entering water temperature	Boolean				

ServiceWaterPiping

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
is_recirculation_loop	Indicates if service water heating piping is a loop and recirculates	Boolean				
are_thermal_losses_modeled	Indicates if thermal losses from the service water heating piping are modeled	Boolean				
insulation_thickness	Pipe insulation thickness	Numeric	m	≥0		From CBECC-Com.
loop_pipe_location	Loop pipe location	<ComponentLocationOptions>				Indicates the type of location for the component but does not indicate if it impacts the simulation, see location_zone. From CBECC-Com.
location_zone	ID of the Zone where the piping is located when IN_ZONE is selected from ComponentLocationOptions	Reference				When present, it indicates that heat transfer into or out of the component turns into a cooling or heating load in the simulation. From CBECC-Com. Constraint to use when implemented :Zone:
length	Pipe length	Numeric	m	≥0		From RESNET
diameter	Pipe section diameter	Numeric	m	≥0		From CBECC-Res.
child	Other service water piping connected to this one as children.	[{ServiceWaterPiping}]				
service_water_heating_design_and_control	Design and control used for service heating piping	{FluidLoopDesignAndControl}				References a FluidLoopDesignAndControl data group which describes how the service water piping is sized and controlled

SolarThermal

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	

Name	Description	Data Type	Units	Range	Req	Notes
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
angle_from_true_north	Solar heater angle from true north, clockwise	Numeric				From CBECC-Com.
solar_savings_fraction	Solar savings fraction	Numeric				Based on ICC-SRCC rating. From CBECC-Com.
collector_area	Solar collector area	Numeric				From CBECC-Com.
collector_type_description	Description of solar collector type	String				From CBECC-Com.
collector_slope	Solar slope from horizontal	Numeric				From CBECC-Com.
tank	Tank that is part of the solar thermal system	{Tank}				Contains a storage tank that is part of the solar thermal system.

ServiceWaterHeatingEquipment

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
heater_fuel_type	Service water heating heater fuel type	<EnergySourceOptions>				
distribution_system	ID of the ServiceWaterHeatingDistributionSystem connected to the ServiceWaterHeatingEquipment	Reference			✓	Data elements linking data groups by reference that don't make sense without the reference are shown as required. Few other data elements than this type and id data elements are shown as required. Other data elements may be required by specific rulesets. Constraint to use when implemented :ServiceWaterHeatingDistributionSystem:
efficiency_metric_types	List of efficiency metrics that correspond to the values in efficiency_metric_values list	[<ServiceWaterHeatingEfficiencyMetricOptions>]				
efficiency_metric_values	List of efficiency metric values that correspond to the descriptors in efficiency_metric_types list	[Numeric]				
draw_pattern	Draw pattern	<DrawPatternOptions>				Draw pattern refers to the water draw profile in the uniform energy factor test. See 10 CFR 430 Appendix E Uniform Test Method for Measuring the Energy Consumption of Water Heaters for more information.
first_hour_rating	First hour rating volume	Numeric	L	≥0		From CBECC-Com.
operating_points	Operating points	[[ServiceWaterHeaterOperatingPoint]]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The power should be directly related to the load. A minimum number of four points is recommended. The ruleset checking specification would normally provide a list operating points that should match what is provided by the simulation.
input_power	Input power	Numeric	W	≥0		From CBECC-Com.
rated_capacity	Rated capacity	Numeric	W			From CBECC-Com.
minimum_capacity	Minimum capacity	Numeric	W	≥0		From CBECC-Com.
recovery_efficiency	Recovery efficiency	Numeric				From CBECC-Com.
setpoint_temperature	Set point temperature	Numeric	C			
compressor_location	Description of where the heat pump for the water heater is located	String				Used when compressor is not located in a specific zone. From CBECC-Com.
compressor_zone	ID of the Zone where the heat pump for the water heater is located	Reference				From CBECC-Com. Constraint to use when implemented :Zone:

Name	Description	Data Type	Units	Range	Req	Notes
compressor_heat_rejection_source	Heat pump heat rejection source	<ComponentLocationOptions>				Indicates the type of location for the component but does not indicate if it impacts the simulation, see compressor_heat_rejection_zone. From CBECC-Res.
compressor_heat_rejection_zone	ID of the Zone for rejecting heat pump heat	Reference				When present, it indicates that heat transfer into or out of the component turns into a cooling or heating load in the simulation. From CBECC-Res. Constraint to use when implemented :Zone:
compressor_capacity_operating_points	Compressor capacity operating points	[[HeatPumpWaterHeaterCapacityOperatingPoint]]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The capacity should be directly related to the evaporator_air_temperature, condenser_water_temperature, evaporator_air_flow, and condenser_water_flow. A minimum number of 64 points is recommended but fewer may be adequate to confirm proper operation especially if the temperatures and flows are not repeated. The ruleset checking specification would normally provide a list list operating points that should match what is provided by the simulation.
compressor_power_operating_points	Compressor power operating points	[[HeatPumpWaterHeaterPowerOperatingPoint]]				Used to confirm the proper operation at various conditions which is based on the nominal value and performance curves, performance maps, or other adjustment mechanism. The power should be directly related to the chilled water evaporator_air_temperature, condenser_water_temperature, evaporator_air_flow, condenser_water_flow, and load on the heat pump water heater. A minimum number of 64 points is recommended but fewer may be adequate to confirm proper operation especially if the temperatures and flows are not repeated. The ruleset checking specification would normally provide a list list operating points that should match what is provided by the simulation.
draft_fan	Draft fan	{Fan}				From CBECC-Com.
has_electrical_ignition	Indicates whether the water heater has electrical ignition	Boolean				From CBECC-Com.
tank	Tank that is part of the service water heating equipment	{Tank}				Contains a storage tank that is part of the service water heating equipment.
status_type	Choice of new, existing, addition, alteration, etc. for each ruleset.	<StatusOptions>				
solar_thermal_systems	Solar thermal systems used for heating service water	[[SolarThermal]]				Contains a list of Solar thermal systems that are part of this service water heating distribution system.
hot_water_loop	ID of the FluidLoop for hot water	Reference				Can be used when heat is supplied to service water heater from a hot water loop or when combination service water and space heating is used. Constraint to use when implemented :FluidLoop:

ServiceWaterHeaterOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
load	Load	Numeric	W			No name and id is needed since typically used as one of a series.
power	power	Numeric	W			The rate of energy used per unit time by the service water heating at the given load.

HeatPumpWaterHeaterCapacityOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
evaporator_air_temperature	Outside dry bulb temperatures of air	Numeric	C			No name and id is needed since used as one of a series.
condenser_water_temperature	Entering condenser temperature of water	Numeric	C			
evaporator_air_flow	Air flow across evaporator	Numeric	L/s			

Name	Description	Data Type	Units	Range	Req	Notes
condenser_water_flow	Water flow across condenser	Numeric	L/s			
capacity	Result	Numeric	W			The capacity available by the heat pump water heater at the given operating point.

HeatPumpWaterHeaterPowerOperatingPoint

Name	Description	Data Type	Units	Range	Req	Notes
evaporator_air_temperature	Outside dry bulb temperatures of air	Numeric	C			No name and id is needed since used as one of a series.
condenser_water_temperature	Entering condenser temperature of water	Numeric	C			
evaporator_air_flow	Air flow across evaporator	Numeric	L/s			
condenser_water_flow	Water flow across condenser	Numeric	L/s			
load	Load	Numeric	W			
power	Result	Numeric	W			The rate of energy used by the heat pump water heater at the given operating point.

Tank

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
storage_capacity	Storage capacity of tank in distribution system	Numeric	L	≥0		From CBECC-Com.
type	Service water heater tank type	<ServiceWaterHeaterTankOptions>				

Name	Description	Data Type	Units	Range	Req	Notes
height	Tank height	Numeric	m	≥0		From CBECC-Com.
interior_insulation	Tank interior insulation R-value	Numeric	K-m2/W	≥0		Insulation that is part of the tank and is inside of the housing. From CBECC-Res.
exterior_insulation	Tank exterior insulation R-value	Numeric	K-m2/W	≥0		A blanket of insulation that surrounds the exterior of the tank. From CBECC-Res.
location	Location	<ComponentLocationOptions>				Indicates the type of location for the component but does not indicate if it impacts the simulation, see location_zone. From CBECC-Res.
location_zone	ID of the Zone containing the Tank	Reference				When present, it indicates that heat transfer into or out of the component turns into a cooling or heating load in the simulation. Constraint to use when implemented :Zone:

ServiceWaterHeatingUse

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
area_type	Service Water Heating Loop Area Type	<ServiceWaterHeatingAreaOptions2019ASHRAE90I>				The enumeration is based on the standard used.
water_serves_type	The use of the water serves the type	<ServiceWaterHeatingFixtureOptions>				
served_by_distribution_system	ID of the ServiceWaterHeatingDistributionSystem that serves this end use	Reference			✓	From CBECC-Res. Constraint to use when implemented :ServiceWaterHeatingDistributionSystem:
use	Usage of service hot water	Numeric				This field together with the use_units and use_multiplier_schedule fields allow detailed description of the ServiceWaterHeating use in the units selected in the use_units field. When describing a flow rate, it indicates the use of hot water from the water heater unmixed with cold tap water.
use_units	Type of units for use of service hot water	<ServiceWaterHeatinguseunitoptions>				
use_multiplier_schedule	ID of the Schedule containing the multiplier for the use of service hot water	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
temperature_at_fixture	Temperature leaving the fixture that is result of mixing hot water from the water heater and cold water from the tap.	Numeric	C			From RESNET
is_heat_recovered_by_drain	Indicates if heat is being recovered from the drain	Boolean				From CBECC-Res.
is_recovered_heat_used_by_cold_side_feed	Indicates if heat is being recovered from the drain is used on the cold side feed	Boolean				From CBECC-Res.

ExteriorLighting

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	The type of exterior lighting fixture	<ExteriorLightingAreaOptions2019ASHRAE901TableG36>				

Name	Description	Data Type	Units	Range	Req	Notes
area	Area of the exterior functional space.	Numeric	m2	>0		The lit area, when the lighting allowance specified by the type is based on area.
length	Linear length measure for exterior functional space	Numeric	m	≥0		For example, used when expressing street frontage or door width
power	Nominal power of exterior lighting fixtures	Numeric	W	>0		
fixture_height	Installation height of exterior fixture	Numeric	m	>0		
is_exempt	Indicates whether the exterior lighting is exempted from requirements	Boolean				
multiplier_schedule	ID of the Schedule containing the multiplier for exterior lighting	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:

RefrigeratedCase

Name	Description	Data Type	Units	Range	Req	Notes
id	Scope-unique reference identifier for instances of this data group	ID			✓	
reporting_name	Descriptive name used in RCT reports if id is not already a descriptive name	String				
notes	Supplementary information to provide context to the model reviewer	String				Notes are not used by the RCT, but may be included in RCT reports to give the reviewer additional information or context
type	Refrigerated case type	<RefrigeratedCaseOptions>				
equipment_category	Equipment Class from referenced standard	<RefrigeratedCaseEquipmentCategoryOptions>				
is_self_contained	Indicates whether unit is self-contained	Boolean				If not self-contained, show as false, and indicates that it has remote condenser
application_temperature	Equipment application temperature	<ApplicationTemperatureOptions>				Based on AHRI 1200
power	Nominal power of refrigerated case	Numeric	W	>0		
power_multiplier_schedule	ID of the Schedule containing the multiplier for refrigerated case power	Reference				If the schedule is not present, the multiplier is always one. Constraint to use when implemented :Schedule:
sensible_fraction	Fraction of energy that is a sensible load on the space.	Numeric		≥-1, ≤1		
heat_gain_fraction	Fraction of energy that is a heat gain to the space.	Numeric		≥-1, ≤1		
volume	Volume of a refrigerated case in cubic meters	Numeric	m3			
total_display_area	Display area of a refrigerated case in square meters	Numeric	m2			
zone	ID of the Zone where the RefrigeratedCase is located	Reference				Constraint to use when implemented :Zone: